

Why Is a Verb Like an Inanimate Object? Grammatical Category and Semantic Category Deficits

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Semantic category effects, such as difficulties in naming animate things relative to inanimate objects, have been explained in terms of the relative weightings of perceptual and functional features within the semantic representations of these concepts. We argue that grammatical category deficits, such as difficulties in naming nouns relative to verbs, can be explained within the same framework. We hypothesize that verb concepts are richer in functional than sensory features and present a model of the semantic representations of animate nouns, inanimate nouns, and verbs. The model demonstrates that sensory feature damage results in a deficit for naming living things but spares verb naming, and functional feature damage results in a deficit for naming inanimate objects and verbs. We then report the assessment results of two patient groups. In accordance with the model's predictions, the "verb spared" patients were consistently worse at naming living things than inanimate objects, and their definitions of both living and nonliving items were lacking in sensory information. We conclude that damage to sensory features in semantic representations causes difficulties in naming concrete nouns relative to action verbs, and within the grammatical category of nouns, animate items will be more severely affected. Imageability was shown to be a strong predictor of naming performance in the "verb deficit" patients, and when this variable was controlled no class effect remained. Production of definitions revealed no differential damage to sensory or functional features, and no consistent effect of animacy in naming was shown. While the model suggests that verb deficits might occur in patients for whom functional features are damaged relative to sensory features, we conclude that the "verb deficit" shown in our patients (and potentially in many previously reported cases) was an artifact of the lower imageability of verbs in confrontation naming tasks. © 2000

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INTRODUCTION

Neuropsychological research has paid much attention in recent years to apparent dissociations in the categories of words which can be selectively impaired or spared after brain damage. There are two main bodies of literature: one investigates deficits specific to grammatical word classes, such as for function versus content words (see Pulvermuller, 1992, pp. 191–192, for a good summary), or for classes within the content word domain, such as for nouns and verbs, summarized in Berndt, Mitchum, Haendiges, and Sandson (1997). The second describes impairments for specific semantic categories, such as animals versus inanimate objects (Hart, Berndt, and Caramazza, 1985; Hillis & Caramazza, 1991; and see Caramazza, 1998, and Grossman, 1998, for summaries of semantic category deficits). Within both of these areas, the question arises as to whether true category-specific deficits occur or whether the phenomenon can be explained by other confounding variables. In both cases this question has not been satisfactorily resolved: evidence has been presented which supports both explanations.

A number of psycholinguistic variables have been shown to affect word retrieval in aphasia. Low frequency words are generally more difficult to retrieve than high frequency (Ellis, Miller, & Sin, 1983; Howard, Patterson, Franklin, Morton, & Orchard-Lisle, 1984; Kay & Ellis, 1987), although a reverse frequency effect has also been reported (Marshall, Pring, Robson, & Chiat, 1998). Similarly, the number of phonemes or syllables in a word can affect retrieval, with either longer words being harder (Howard et al., 1984) or easier (Best, 1995) to retrieve. Imageability (the ease with which a word produces a mental image) is necessarily correlated with concreteness, but they are not exactly the same thing. Items rated lower in imageability (and hence usually more abstract) are less well retrieved in aphasia (Franklin, Howard, & Patterson, 1995; Nickels & Howard, 1995), but again the reverse effect has been reported both in semantic dementia (Breedin, Saffran, & Coslett, 1994) and aphasia (Marshall, Chiat, Robson, & Pring, 1996).

This paper draws strands of both grammatical and semantic category deficits together in a study of a small case series and suggests that the same phenomenon might account for both types of category effects in aphasia. We make two main assertions. The first is that many reported “grammatical class-specific deficits” are not truly class specific, but the result of the confounding of class with imageability. The second assertion is that “true” word class-specific deficits are in fact due to differences in the distributions of semantic feature types, analogous with the favored explanation of semantic category deficits. We suggest that pictureable objects have a greater weighting of sensory features compared with actions, which are primarily

motor and functional. Thus, in general, concrete nouns have more perceptual features, and verbs (as well as abstract nouns) more functional.

Semantic Category-Specific Deficits

Many reports of category-specific disorders have not controlled sufficiently for confounding variables, as pointed out by Funnell and Sheridan (1992) and Stewart, Parkin, and Hunkin (1992). However, cases of category-specific deficits do exist that are robust even when all potential confounds are scrupulously controlled (Lambon Ralph, Howard, Nightingale, & Ellis, 1998; Howard, Best, Bruce, & Gatehouse, 1995). These two studies appear to be the only ones in which imageability has been controlled (or entered into the regression analyses). The variables which are most confounded with animacy are frequency and familiarity, so other studies have tended not to be concerned with imageability.

Category-specific semantic deficits have been explained in various ways. One proposal is that the semantic store might be categorically organized, with (at least partially) separate meaning representations for categories, such as animals, fruit, and vegetables (Hart et al., 1985). A second suggestion is that animate and inanimate items may differ because of differences in the proportions of shared features. McRae, de Sa, and Seidenberg (1997) argue that animate items have greater clustering of semantic features: for example, many animals share the same features of eyes, ears, nose, fur, four legs, and so on. Thus there is much overlap, and the retrieval of a single item might be impeded by having so many similar semantic competitors. A third explanation is that the semantic representations of animals and man-made items differ primarily in terms of the weighting of sensory and functional information contained in their representations, such that the semantics of animals, birds, fruit, and vegetables is heavily based on the senses, particularly vision (Warrington & Shallice, 1984), while for artifacts function is more important.

There are difficulties with each of these accounts. First, that the semantic store is subdivided into categories is contentious. Reports exist of patients who exhibit a deficit specifically for animals (Hillis & Caramazza, 1991; Caramazza & Shelton, 1998), but also those for other classes of "animate" items, including fruit and vegetables (Hart, et al., 1985; Farah & Wallace, 1992). Such reports might suggest that a distinction does not lie on an animate-inanimate dimension, but that more specific categories are involved (although Farah and Wallace noted some difficulty in their patient's naming of other animate items, plants, animals, and insects, they did not report a full investigation of these other categories). Patients with deficits for inanimate items also tend to be impaired at body parts (technically not inanimate) and yet might be spared at musical instruments (Gainotti & Silveri, 1996). Warrington and Shallice (1984) also reported a patient for whom the semantic representations of living things were degraded but whose knowledge of

body parts was spared. These trends would seem to support the sensory–functional distinction more than that of specific semantic categories or a simple animate–inanimate distinction. Musical instruments, for example, are inanimate, yet, at least for nonmusicians, distinguished primarily in terms of their appearance; as a result difficulty with musical instruments co-occurs with a difficulty for living things.

Categories such as fruit and vegetables, which contain many exemplars with similar perceptual structures, have been shown to be named more slowly in experiments on non-brain-damaged subjects (Lloyd-Jones & Humphreys, 1997). This evidence supports the second explanation in terms of the clustering of semantic features. Lloyd Jones and Humphries argue that their results should generalize to other categories comprising visually similar items. One might assume that the category of animals also contains many structurally similar items, and picture naming would be slower in normals (and more impaired in brain-damaged individuals) for animates than inanimates.

The last explanation, which explains differences between categories in terms of differences in the importance of functional and sensory features can easily be extended to explain both grammatical category effects and patients' performance with the "exceptional" categories, such as musical instruments and body parts. Gainotti and Silveri (1996) point out that the "'living/non-living' dichotomy suffers from important, systematic exceptions" and argue that animate items (except body parts) are distinguished primarily in terms of sensory features, whereas inanimate items (apart from food, musical instruments, and geographical features) are distinguished more in terms of their "functional" properties. Caramazza (1998) provides an excellent summary of semantic category-specific cases in the literature and the merits of the various explanations. Most crucial is his table of major dissociations reported to date between nonliving, "living inanimates" (e.g., fruit and vegetables) and "living animates" (animals). There have been reports of five of six possible combinations of deficits of these three categories. The one that has not been reported is an impairment of both animals and nonliving things, with spared living inanimates. Assuming that food items or living inanimates fall midway between the other categories in terms of their relative weighting of sensory and functional features, there would be a sensory–functional "gradient" from living animates to living inanimates and then to nonliving things. If this is indeed the case, the one dissociation we should not expect is impairment of opposite ends of the gradient but not the category in the center, and this is the very combination not reported.

The feature clustering explanation is not incompatible with the sensory–functional theory: categories comprising exemplars which are similar in perceptual structure are those for which sensory information is key. The categories used in experiments by Lloyd-Jones and colleagues which exemplify the lack of structurally similar competitors are inanimate, such as clothes and furniture. In this way the "clustering" hypothesis and the "sensory–

functional'' theory complement each other. The fact that animate concepts have more features which intercorrelate (and most of these are sensory) might lie behind the prevalence in the literature of deficits for living things, with the reverse being much less common. Alternatively, this imbalance might suggest that sensory features have more salience in semantic memory or are less distributed around the cortex than functional features and hence more prone to damage by localized lesions.

If semantic feature correlations and the sensory–functional continuum of semantic space are both at work in apparent category-specific deficits, we still need to explain the reported dissociations between the animate categories of animals versus fruit and vegetables. This could again lie in the types of features, but more specifically the type of sensory features, which correlate among category exemplars. Lloyd-Jones and Humphreys (1997) selected the category of fruit and vegetables because it was the category with the highest degree of visual similarity in the Snodgrass and Vanderwart (1980) pictures in terms of contour overlap. If they had selected the set of animal pictures, the contour overlap between items would not be as great; nevertheless, other kinds of sensory features do overlap within in the category of animals: fur, eyes, nose, number of legs, and so on. Features such as these are also visual, but they do not pertain as closely to the outline, or overall shape, of an item. They are also, perhaps, more strongly based on factual knowledge about the bodies of animals that we might never have experienced at first hand, but only in pictures or on the television. Russell (1914) discussed the difference between knowledge gained by direct sensory input and that gained only by description, noting that most of the “facts” we know about the world are things of which we have no direct experience. He considered knowledge by experience to be more fundamental: “when I actually see Memorial Hall, even if I do not know that that is its name, and even if I make no propositions about it, I must be said to know it in some sense more fundamental than any which can be constituted by the belief in true propositions describing it” (Russell, 1914, p. 154). While information based on knowledge should probably lie on the functional end of the continuum, this information is still based on visual input from pictures and so on and is therefore sensory. It can be argued, however, that the type of sensory features which are central to fruit and vegetables are not the same in nature as those which are central to animals. The feature data collected by Devlin, Gonnerman, Andersen, and Seidenberg (1998) for their connectionist model showed a slightly higher ratio of sensory to functional features for animals compared with fruit and vegetables, but also that fruit and vegetables have more intercorrelating features. Devlin and colleagues demonstrated that a model based on differing distributions of sensory and functional features could produce semantic category effects both with focal and with widespread semantic damage. These authors also note that there are no reported cases of selective damage to any specific category within the category of inanimate objects. This in turn casts

doubt on the assertion that “category-specific” deficits actually reflect true categories within the semantic system.

The semantic category-specific literature contains many examples of evidence supporting the theory that category-specific deficits are directly related to differential damage to sensory or functional features in semantics. We can find only two exceptions. Samson, Pillon, and De Wilde (1998) contend that their patient Jennifer (who had a living things deficit) had no deficit for visual attributes. She was, however, below the normal range on the BORB object decision test (Riddoch & Humphries, 1992) and she was reported to have a “visual long-term memory deficit.” These visual problems might have contributed to the apparent deficit for living things. Lambon Ralph et al. (1998) describe two contrasting cases: DB was impaired at living things, but showed no difference between visual and functional knowledge; IW had poor knowledge of visual attributes and yet was better at naming animates than objects. While this counterevidence to our assertion cannot be dismissed, DB was very poor at the color test developed by Graham (1995; Graham, Becker, Patterson, & Hodges, 1997), in which the subject is required to color black and white line drawings appropriately. This suggests that visual semantic representations were severely impaired. IW’s “animacy” effect was not apparent on any individual controlled test, but only by conducting regression analyses including the many other psycholinguistic variables which predict her naming success. Furthermore, according to her own report, premorbidly she was very fond of animals and used to visit zoos often (personal communication with the patient).¹

Grammatical Category-Specific Deficits

The aphasia literature abounds with cases of patients who apparently find retrieval of verbs more difficult than that of nouns (Caramazza & Hillis, 1991; Jonkers & Bastiaanse, 1998; Zingeser & Berndt, 1988; Orpwood & Warrington, 1995; Rapp & Caramazza, 1998). Many of these patients are nonfluent and agrammatic (Zingeser & Berndt, 1990; Miceli, Silveri, Villa, & Caramazza, 1984). Fluent anomic patients are reported to be impaired in the retrieval of nouns, but in most cases their verb retrieval is not formally tested (Kay & Ellis, 1987). The spontaneous speech of fluent anomic patients often lacks nouns, but seems to have sufficient verbs to allow grammatical speech production. It has been demonstrated that an inability to produce anything but high frequency words would result in this pattern (Marshall, 1977; Bird, Lambon Ralph, Patterson, & Hodges, 2000). In stud-

¹ It is possible, then, to argue that IW’s relatively preserved animate categories are related to greater familiarity. There is at least one difficulty with this, however. She was, for many years, a professional florist and continued to be an enthusiastic gardener and flower-arranger. She was, however, essentially unable to name flowers of any kind, including the most common exemplars such as “rose” or “daffodil.”

ies comparing noun and verb production in single word naming, frequency and length are normally controlled, but not imageability (Hillis & Caramazza, 1995; Miozzo, Soardi, & Cappa, 1994; Berndt et al., 1997; Bastiaanse & Jonkers, 1998). It can be assumed that all items used in confrontation naming must necessarily be quite highly imageable, but this does not mean that items are equally imageable. We will return to this problem later.

Caramazza and Hillis (1991) described two aphasic patients, HW and SJD, both with impaired verb production, but restricted principally to verbs in oral and written production, respectively. The authors interpreted this dissociation as evidence that grammatical class deficits arise at a lexical level, and the word classes are represented redundantly in phonological and orthographic output lexicons and not in the semantic system. Gainotti, Silveri, Daniele, and Giustolisi (1995), however, in their review of grammatical category deficits in the literature, cite data from patients (GG and RA in Daniele, Silveri, Giustolisi, & Gainotti, 1993; Daniele, Giustolisi, Silveri, Colosimo, & Gainotti, 1994) exhibiting verb deficits due to degenerative disease which strongly contradict this hypothesis: as their disease progressed, the patients' early deficits limited only to production (GG), and limited only to one output modality (RA, as shown in HW and SJD), emerge in comprehension and in the other output modality, respectively. Gainotti et al. (1995) suggest that "a basic semantic disorder may have different levels of clinical expression in the various lexical modalities" (pp. 251–252) and also that a mild semantic impairment might at first only be manifested in stringent production tasks "owing to their greater demands as compared to comprehension."

Marshall et al. (1996) describe a patient, RG, who was more impaired in the production of nouns than verbs and better at abstract than concrete nouns. In spite of the relative sparing of verbs relative to nouns, this patient's verb comprehension demonstrated some dissociations in performance according to the types of semantic relationships within verb concepts. RG was more impaired in comprehension when the difference between target and distracter was in manner (he would, for example, select the distracter *crawl* for the target *slide*), which they deem to be a perceptual feature. RG was, however, spared in the understanding of thematic roles, a functional feature of a verb (so he would have no difficulties in distinguishing *give* from *take* and *buy* from *sell*). Marshall et al. (1996) suggest the same underlying semantic deficit influences spared production of abstract nouns in relation to concrete and sparing of thematic role information in contrast to manner of action. This explanation for a grammatical category deficit is akin to the sensory/functional feature hypothesis of semantic category deficits.

Converging evidence for a semantic basis to grammatical category effects also comes from the literature on optic aphasia, a naming disorder in which inferior performance is shown on naming visual stimuli than tactile stimuli or naming to definition. Druks and Shallice (1996) describe a patient, LEW,

who was equally impaired at nouns and verbs in visual naming, but his performance on verbs improved both when his body was manipulated to act out verbs and when he was required to “pantomime” the pictured action. Those he could unambiguously perform, he could then successfully name. Ferreira, Giusiano, Ceccaldi, and Poncet (1997a) report on CN, who, unlike LEW, could name verbs from pictures better than nouns. His noun naming was improved by viewing gestures depicting the use of the objects to be named. These authors suggest the existence of functional and visual semantic systems. Certainly the additional of information other than visual aided the naming performance of both these patients. This, however, does not necessitate the existence of two separate systems. The difference between the semantic representations of nouns and verbs might be similar to the difference suggested between animate and inanimate objects: the former (in both cases) depends more on sensory than functional information. In a distributed semantic system, it is possible, in the patients just described, that the area of their semantic system which is most damaged is that which contains many sensory features or the pathways from those features to phonological and/or orthographic representations.

Studies Which Have Considered Both Semantic and Grammatical Categories

Recently Moss, Tyler, Durrant-Peatfield, and Bunn (1998a) examined the naming and semantic knowledge of a patient, RC, who had a semantic-category deficit for living things following herpes simplex encephalitis. In noun naming, he named artifacts with greater accuracy than living things and showed better knowledge of properties shared between category exemplars (e.g., *has ears*, *breathes*) than distinctive, item-specific properties (e.g., *has stripes*) within the living domain. Moss, De Mornay Davies, Jeppeson, McLellan, and Tyler (1998b) had also examined RC's semantic knowledge of verbs in property verification and definition production. This was also impaired, but significantly more so for verbs which denoted actions performed by living things which were specific (such as *gallop* and *peck*), but not general (such as *eat* and *sleep*). This shows the same pattern as for the semantic representations of nouns and prompts Moss et al. (1998a) to suggest that “conceptual representations of nouns and verbs are captured within a single distributed semantic system, rather than independent, localized stores” (p. 95). The properties of verbs that are described as “specific” and those of nouns which are described as “distinctive” in these studies are those which depend greatly on sensory features, and it was these properties which were most degraded in RC. Conversely “shared” and “general” features of nouns and verbs, respectively, were relatively spared, and we would suggest that RC's sparing of these more functional features caused artifacts to be better named than living things.

The majority of studies of category-specific deficits consider either semantic or grammatical categories, and very rarely are both types investigated simultaneously. There are a few examples of patients described in the literature who are poor at naming animates and also spared in action naming. Ferreira, Giusiano, and Poncet (1997b) described three patients with selective impairments for naming animals in comparison with actions and tools. It was noted in Miozzo et al. (1994) that AL, who was better at naming verbs than nouns, was also better at naming inanimate than animate nouns. The authors suggested this might be due to the imbalance of familiarity in this picture set and did not pursue the relationship further. Shelton, Fouch, and Caramazza's (1998) patient IOC is described as having a selective sparing of body part naming. Such cases have usually been accompanied by a deficit for animate objects, and relative sparing of nonliving things, a combination which we see as evidence for the functional-sensory explanation rather than true category specificity. Closer examination of their data reveals that indeed, inanimate items are relatively spared compared with animate items: if we remove body parts (on the basis that they are animate but do not associate with animates in category-specific deficits) and musical instruments (on the basis that they are inanimate, but associate with animates in category-specific deficits, presumably due to their preponderance of sensory features), then what emerges is a greater deficit for the living than the nonliving items. IOC also had spared verb naming, but this was not further investigated. If the category-specific deficit was indeed due to damage to the sensory featural representations, then this, along with the other animate-noun and inanimate-verb cooccurrences described above, lends further support for the assertion that grammatical class deficits might be due to similar semantic factors to semantic category deficits.

Magnie, Ferreira, Giusiano, and Poncet (1999) report a case of category-specific object agnosia. The patient (JMC) was able to recognize and gesture manipulation of many inanimate objects, but no animals, fruits, vegetables, or musical instruments. His ability to give appropriate definitions was also limited to those categories for which he could demonstrate the items' use, and he could name only items for which he could provide a definition of use. He was, however, good at naming actions, whether they involved the use of an object or not. This leads the authors to suggest an important role for sensorimotor experience in object recognition and naming. In contrast, they say, most animals do not evoke any gestures, and those for fruit and vegetables are little more than simple gripping. The same argument applies to musical instruments, if the individual had no experience of playing an instrument, and reportedly JMC had never played a musical instrument. Magnie and colleagues suggest that the preservation of action identification aided JMC in recognizing and naming objects. It is possible, however, that the preservation of knowledge about actions was linked to their lack of dependence on sensory/visual information. Those items (animates and musical

instruments) about which JMC had predominantly only perceptual knowledge might have been damaged because those types of semantic features had been selectively damaged.

Reports of reverse animacy effects are less frequent, and we have found none that investigate verb production. Sacchet and Humphreys (1992) report a category-specific deficit for artifacts exhibited by their patient CW. They did not describe the patient's verb production, but the speech sample they provide shows a marked lack of verbs. The co-occurrences of verb deficits with inanimate deficits and noun deficits with animate deficits are rarely reported, but no contradictory evidence has so far emerged.

Why a Verb Is Like an Inanimate Object

In a survey of neuroanatomical correlates of category-specific disorders, Gainotti et al. (1995) suggest that different categories of knowledge might be located at specific brain areas linked with the perceptual and motor brain mechanisms needed to acquire these concepts. They find that patients reported to have a verb deficit tend to have lesions involving anterosuperior portions of the left temporal lobe and the inferoposterior parts of the left frontal lobe, whereas deficits in the production of nouns correlate with lesions of the lateral and inferior parts of the temporal lobe which extend toward the posterior association areas. They refer to the Geschwind (1967) model suggesting that sensory semantic information might converge in posterior association areas and that this is why object names are impaired with such lesions. Similarly, action names, for which semantic representations must depend greatly on motor schemata, suffer when more anterior portions of the brain are lesioned. Gainotti et al. (1995) suggest therefore that the differing features of the semantic representations of verbs and nouns lie behind their differential impairment found in brain damage and that this correlates with areas of the brain known to be associated with motor and sensory functions.

The Gainotti et al. (1995) survey of deficits for living and nonliving categories shows quite a consistent pattern of lesions also involving the inferior part of the temporal lobes in cases of impairment of living things (but IW, the Lambon Ralph et al. (1998) patient is an exception). There is, therefore, an area of the brain which is affected both in deficits of noun production (with spared verbs) and semantic deficits for living things, which suggests that both phenomena can be explained by damage to part of the semantic system which "may play a critical role in processing, storing and retrieving the semantic representations of these semantic categories" (Gainotti et al., 1995, pp. 259–260). Category-specific disorders for inanimate items correlate with lesions involving frontoparietal areas, which correspond with somatosensory and motor functions. We have already seen that verb deficits too are associated with more frontal lesions. Deficits for inanimate objects,

such as tools, are often associated with a corresponding deficit for body parts. The most important semantic feature of tools and body parts is surely what one does with them: body parts particularly involve movement and tools involved manipulation and use (a functional rather than sensory weighting). In other words, their actions and how they are acted upon are the most salient parts of their meaning. Herein lies the similarity of verbal function and semantic content of verbs and man-made items and body parts. It should be no surprise, therefore, that similar areas of cortex should be associated with these items and that patients for whom naming of inanimate objects is spared relative to animates should also have action naming spared relative to objects in general.

The first section of this paper describes an attempt to model semantic space in terms of sensory and functional feature representations. Representations of "animate nouns," "inanimate nouns," and "verbs" are produced simply by varying the distributions of these types of features. Four models are presented, with differences in design and in the assumptions of the actual ratio of sensory to functional features. Each is tested by selectively damaging feature types, and the resulting patterns of deficits in word naming are described. We demonstrate that the same basic predictions arise from each of these models: deficits for inanimate objects should be accompanied by poor verb naming, and sparing of verb naming should co-occur with sparing of the production of inanimate objects relative to living things.

We go on to investigate these predictions in two patient groups, "verb deficit" and "verb spared." First we examine the robustness of the noun-verb effects using abstract as well as pictureable items. These results suggest that many apparent verb deficits are in fact due to imageability effects: when imageability is controlled, performance on nouns and verbs is equal. Then we compare the patient groups' performance in naming animate and inanimate nouns, to demonstrate that the "verb spared" group are indeed worse at naming animate nouns. Finally we test these patients on definition production and show that, as predicted, the "verb spared, inanimate spared" group suffer a deficit for sensory feature knowledge. We will conclude that this sensory feature deficit is the cause of both their animacy effect and the grammatical class effect shown in picture naming.

MODELING ANIMATE NOUNS, INANIMATE NOUNS, AND ACTION VERBS IN SEMANTIC SPACE

In this section we describe four different "models" of semantic space, which includes concrete nouns, both animate and inanimate, and action verbs. We then "lesion" the models and show the category effects which emerge and the predictions that arise. The modeling is based on the simple premise that these types of words vary in the distribution of feature types, and we demonstrate how this can result in consistent predictions when the models are lesioned.

Caveats

The models assume all things are equal except those factors manipulated. This includes variables such as frequency, length, age of acquisition, familiarity, and visual complexity. It also assumes in "testing" that a set comprising equal numbers of animate and inanimate nouns is tested along with verbs and that each set is controlled for all variables (except imageability, which is considered separately). In real life, it would be impossible to produce such large numbers of controlled items. Verb and noun sets used in confrontation naming do not normally have nouns controlled for animacy, nor is imageability controlled across the grammatical classes. Most animacy-controlled sets are not controlled for all the relevant variables, as we discussed in the Introduction. Clustering of semantic features, which has been suggested as another factor in category-specific deficits, is not taken into account here.

There is obviously more to semantic representations than sensory and functional features: abstract words probably have little of either, but their meaning is more dependent on verbal definitions. This might best be represented on a third "abstractness" axis, but this is not considered here, in order to keep the models as simple as possible. Within sensory and functional features there probably exist many subsets. For simplicity the model subsumes features related to vision and hearing, for example, under the term "sensory."

The use of such broad terms as animate/living things and inanimate/non-living things/artifacts is also a gross generalization. Data from brain-damaged individuals suggest that deficits for inanimate things also tend to involve body parts, and those for animate things also involve musical instruments. This, as we have already suggested, lends further support for the theory that the underlying deficit is due to damage to types of semantic features, rather than to the specific semantic category. For ease of reference, therefore, we use the terms "animate" to include animals, fruit and vegetables, and other nouns which we believe are heavily dependant on sensory features (such as musical instruments). The term "inanimate" would probably include body parts and perhaps some food items. When these terms are used in the modeling, we refer to concrete nouns which depend to different degrees on sensory and functional features.

Distributions of Semantic Feature Types

Studies have varied greatly in the manner in which they have collected data to support the animate/inanimate distinction in terms of featural representations. Farah and McClelland (1991) used the method of underlining features in definitions taken from a dictionary. Hodges, Patterson, Graham, and Dawson (1996) asked subjects to describe each item as if for a person who had no idea what it was. Subjects had 1 min for each item, and after practice items, feedback was given with further examples of the type of sen-

sory and functional information that might be included. McRae et al. (1997) gave subjects 2 min per item and asked them to give a written list of features for each object. They asked for different feature types "physical (perceptual) properties . . . functional properties . . . and encyclopaedic facts." Space on the sheet was provided for 10 of each feature type. The results of each of these studies are shown in Table 1. The final column shows their ratios converted to the number of sensory features for each single functional feature, so that they can all be viewed in the same way.

In the first two studies, in which subjects were not constrained to give equal numbers of sensory and functional features, animates received more features in total than did inanimates. Both animates and inanimates received more sensory than functional features, but the difference is greater for animates. This shows that animates are given a larger proportion of sensory features than are inanimates. The actual ratios vary enormously across the studies: Farah and McClelland (1991) showed a very large difference in the weighting between the two semantic categories; Hodges et al. (1996) found very little difference. For our modeling we calculated an average across the three studies to give a starting point for the distribution of sensory and functional features. This average ratio was 3.53:1.00 for animates and 1.32:1.00 for inanimates.

Imageability

Imageability has been consistently shown to affect the production of patients (Franklin et al., 1995; Nickels & Howard, 1995) and indeed normal subjects (Strain, Patterson, & Seidenberg, 1995): generally speaking, lower imageability items are slower or less likely to be accessed. It is thought to be most relevant to the semantic level and hence might have an important effect on category deficits. Howard et al. (1995) obtained imageability ratings for animate and inanimate items and found that the latter tended to be rated slightly lower. The mean rating for animates was 600 and for inanimates 579. Ratings for pictureable action verbs, collected for the verb and noun naming test which we describe later, average 459.

Verbs attract considerably lower imageability ratings than nouns (considering only those suitable for naming tests). Inanimates tend to be only slightly lower in imageability than animates. A patient who shows an imageability effect, such that lower rated items are less likely to be produced, might therefore be expected to perform worse at verbs than nouns. An effect of animacy (inanimates less likely to be named) is unlikely to reach significance, but a trend might be seen across patients with strong imageability effects. It is possible to create lists of animate and inanimate nouns controlled for imageability, as they have considerable overlap, but this is not the case for pictureable nouns and verbs. Using the model, however, we can build in sufficient variability within each word class to allow overlap and hence use controlled subsets of nouns and verbs.

TABLE 1
Ratios of Sensory to Functional Features Obtained in Three Studies

Reference	Animate		Inanimate		Animate sensory: functional	Inanimate sensory: functional
	Sensory	Functional	Sensory	Functional		
Farah and McClelland (1991)	2.68	0.35	1.57	1.11	7.70:1.00	1.40:1.00
Hodges et al. (1996)	5.30	4.20	4.20	3.40	1.26:1.00	1.24:1.00
McRae et al. (1997) ^a	758	469	852	854	1.62:1.00	1.00:1.00

^a These authors analyzed the data in terms of feature frequencies, and here ‘functional’ includes all information other than that which they describe as “aspects of a concrete entity”; hence it incorporates classification and related cognitions.

What in the cognitive system is imageability tapping? The instructions for eliciting these ratings focus on sensory information, which is used to “produce a mental image” from a word. It is reasonable to assume, therefore, that these ratings are some measure of the number of sensory features in the semantic representation of each item. The difference in average rated imageability between animate (mean rating 600) and inanimate (mean rating 579) items is very small. The mean difference between the number of sensory features given for animate and inanimate items in studies of features from their definitions (see Table 1) is, however, very much greater.

We saw in the studies investigating distributions of sensory and functional features that animates obtained a greater number of features in total than did inanimates (McRae et al. (1997) is not included as they controlled the number of features, and their method of analysis is also different). Averaging across the other two studies, the total number of features given by subjects is 6.27 for animates and 5.14 for inanimates.

These differences in the total number of features is in the same direction as for sensory features (more for animates), but is much less pronounced. The difference is still greater than that seen in the mean imageability ratings for animate versus inanimate items, but it is closer. Imageability might be related to sensory features or to the total number of features (the more the sensory features, the more the total features, and the higher the imageability) or both. Animacy effects observed in patients can be eliminated when imageability is controlled (Howard et al., 1995), but not in all cases (Lambon Ralph et al., 1998). If animacy effects are due to the number of sensory features which are available in a damaged system, and imageability is merely a measure of the number of sensory features, one might predict that animacy effects should disappear when imageability is controlled. This, however, is not the case. We suggest therefore that imageability more closely reflects the total number of semantic features available in a semantic representation, a kind of measure of semantic richness. Plaut and Shallice (1993) discussed the idea of semantic richness. These authors note the correlation between imageability and ease of predication and that ease of predication maps exactly onto the relative difficulty of producing different parts of speech in patients with deep dyslexia. Thus verbs, which are of lower imageability, are less often produced than concrete nouns. In their connectionist model, Plaut and Shallice (1993) represented this difference by using fewer features for words which were more abstract, and an effect of imageability observed in a patient or a normal subject is a demonstration of a threshold that occurs normally when we try to access a word.

In our modeling, we make the assumption that a particular threshold of activation must be reached to attain production of the target word. By this we mean that a particular number of features must be activated to achieve naming. The probability of successful (and quick) access to a word is increased by having a greater number of semantic features associated with that

semantic representation. If lower imageability words do have fewer features, then imageability can be built into the model by giving the representations differing numbers of features. Morrison, Ellis, and Quinlan (1992) found no independent effect of imageability in picture naming (nouns only), when age of acquisition was included in the regression analysis, but as this included only concrete objects, the range of imageability could not have been very wide. Barry, Morrison, and Ellis (1997) found an effect of imageability on normal subjects' picture naming latencies, which did or did not reach significance depending on the word frequency count used in the analysis. Lower imageability items are produced more slowly by normal subjects in word naming (Strain et al., 1995) when the words are low frequency and have irregular spellings. These authors propose that when the process of translating orthography to phonology is slow or "noisy," a semantic factor such as imageability takes its effect. This idea can be applied to confrontation naming when brain damage caused, for example, by CVA results in "noisy" access. In effect, the threshold is then raised, and items with greater semantic richness reach that threshold more easily. In a patient we will see the normal pattern, but often greatly exaggerated, so that abstract items (unless perhaps they are very high frequency) will be much less likely to be produced, while items rich in semantic features (high imageability) are successfully named.

Four different models are described which are distinguished by the manner in which imageability is represented in the system and by the weightings of features types. For each model we will ask the following questions: 1. Does the model show an association such that a deficit for living things is accompanied by a deficit for nouns relative to verbs? 2. Does the model show an association such that a deficit for nonliving things is accompanied by a deficit for verbs? 3. Does the model produce animacy effects as reported in both directions? 4. Is this robust under control for imageability?

The first two models (A and B) assume the mean total number of features is equal across the three types of words and that imageability is represented in the number of sensory features. The latter two models (C and D) assume differences in the mean total number of features across word types and that these differences represent semantic richness, or imageability. The difference between model A and model B is in the distribution of feature types: for the two categories of nouns, model A uses the mean ratios derived from the three studies described above, and model B uses imageability ratings to represent the actual number of sensory features. The ratio of sensory to functional features for verbs is hypothetical, based on the assumption that verbs have fewer sensory than functional features. In subsequent models, the actual mean imageability ratings of verbs from the verb and noun naming test (described later) are used. Models C and D both use imageability ratings to derive the total number of features; again the difference between the models lies in the distribution across sensory and functional. Model C uses the same mean ratios as in model A to derive the distribution of types of features, and

TABLE 2
Number of Features and Ratios in Each of the Four Models

Model and word type	Total features	Sensory features	Functional features	Ratio of sensory: functional features
A				
Animate	1000	780	220	3.53:1.00
Inanimate	1000	570	430	1.32:1.00
Verbs	1000	400	600	0.68:1.00
B				
Animate	1000	600	400	1.50:1.00
Inanimate	1000	579	421	1.38:1.00
Verbs	1000	459	541	0.85:1.00
C				
Animate	600	468	132	3.54:1.00
Inanimate	579	329	250	1.32:1.00
Verbs	459	184	275	0.67:1.00
D				
Animate	600	340	260	1.31:1.00
Inanimate	579	319	260	1.23:1.00
Verbs	459	199	260	0.77:1.00

model D keeps the total number of functional features constant. The actual numbers of features and ratios are shown in Table 2. The ratios of sensory to functional features in model D for animates and inanimates are very close to those obtained by Hodges et al. (1996), which represent the most conservative estimate of the difference in the distributions across the semantic categories.

How the Models Simulate Naming

Microsoft Excel 7.0 was used to create and run the models. Semantic representations were created by generating random numbers with the mean and standard deviations specified (standard deviations for the total number of features were always approximately 20% of the mean). Four hundred exemplars of each category of word were produced in each model: each "word" comprised one cell containing the number of sensory features and one containing functional features. To simulate naming, the total of these two cells had to be equal or above a set threshold. In the "normal" (prelesion) setting, the threshold was set so that approximately 98% naming was achieved. The "imageability" (total sensory features in model A and B and total features in models C and D) of the correctly named items and the errors was calculated. In all cases, the correctly named items were of a higher mean imageability than the errors. This was an automatic result of the correlation between the number of sensory features and the total features. This feature was not built into the models, but emerged because items with more sensory

features generally had a greater total number of features, and the total number of features predicted naming success.

The models were “lesioned” by decreasing the number of sensory and/or functional features by a set proportion, to simulate damage to sensory and/or functional features in semantic representations. Different levels of “lesion” were tested and the resulting proportions of the three word types which were successfully named were calculated. We refer to the different levels of damage as mild (40% of features removed), moderate (70% of features removed), and severe (90% of features removed). To look at the effects on “imageability controlled subsets,” in models A and B the subset of items which had (prelesion) a number of sensory features within a set range were examined; in models C and D we examined a subset of items which had (prelesion) a set range of the number of total features. In models C and D we also investigated the effects of raising the threshold of total features required for naming. This was not relevant for models A and B, as the total features were constant across the three word types.

Results

Figures 1–8 show the results for each of the four models, first without and then with control for “imageability.” For each of the four models, we can answer yes to both questions 1 and 2: there are associations shown between performance on living things and noun deficits and between non-living and verb deficits. Only in models A and C, however, are the animacy effects significant, so we can answer yes to question 3 for these models, but not for B and D. B and D fail to replicate the animacy effects reported in the literature because there is insufficient difference between the feature ratios for animate and inanimate things. Only model C provides an affirmative to question 4 (are the animacy effects robust under control for imageability?) (Figs. 5 & 6).

As models B and D failed to show the required animacy effects, the remainder of the discussion will focus on models A and C. In both these models (which utilize the same feature ratios), a much more severe lesion was required to get a reverse animacy effect than was needed to obtain the animacy effect in the usual direction. This is because inanimate concepts have a more even distribution of the two types of features, so if only one type is damaged, many other features are spared. In contrast, animate concepts depend greatly on sensory features, so when they are damaged, they have few functional features to “fall back on.” When inanimates were significantly worse than animates, a very severe verb deficit was present, because the reverse animacy effect was due to severe damage to functional features, on which verb concepts are more dependent. Combined damage to sensory and functional features resulted in similar effects, depending on which feature type was damaged more. Greater sensory damage makes animates and nouns less likely

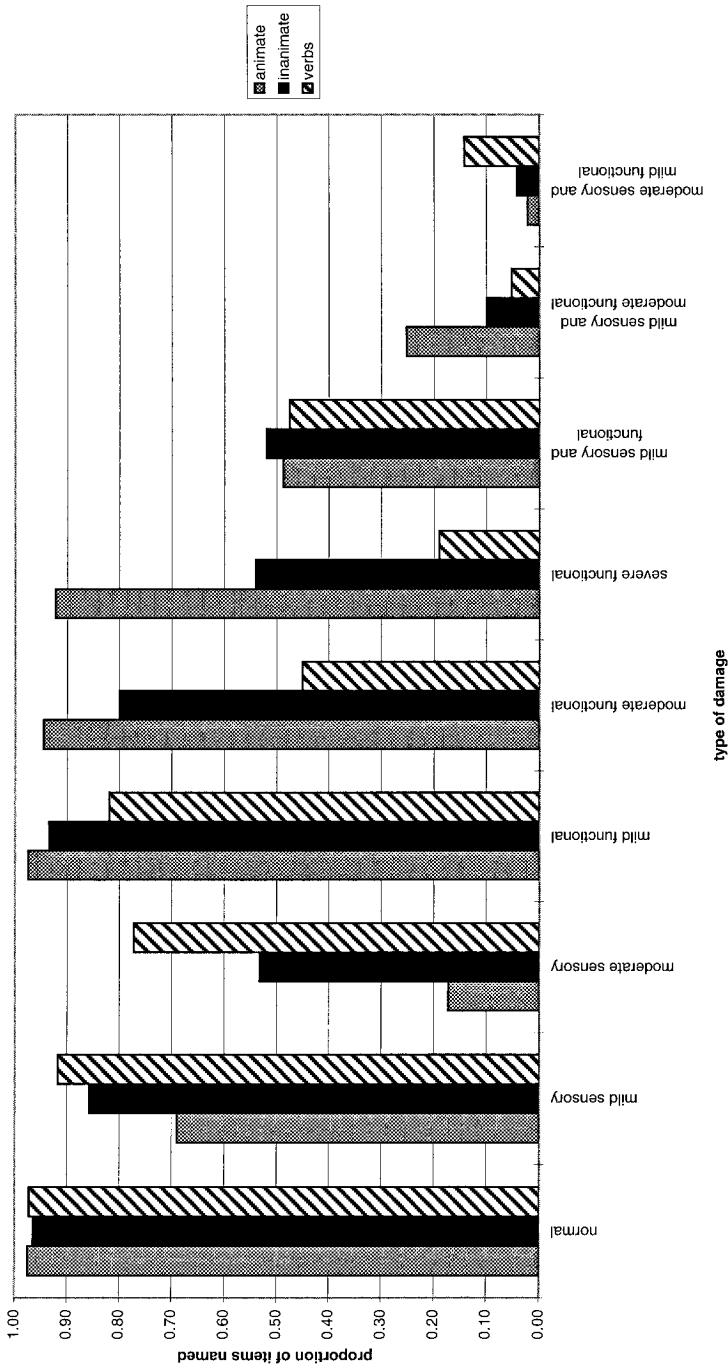


FIG. 1. Model A: Effects of different lesions.

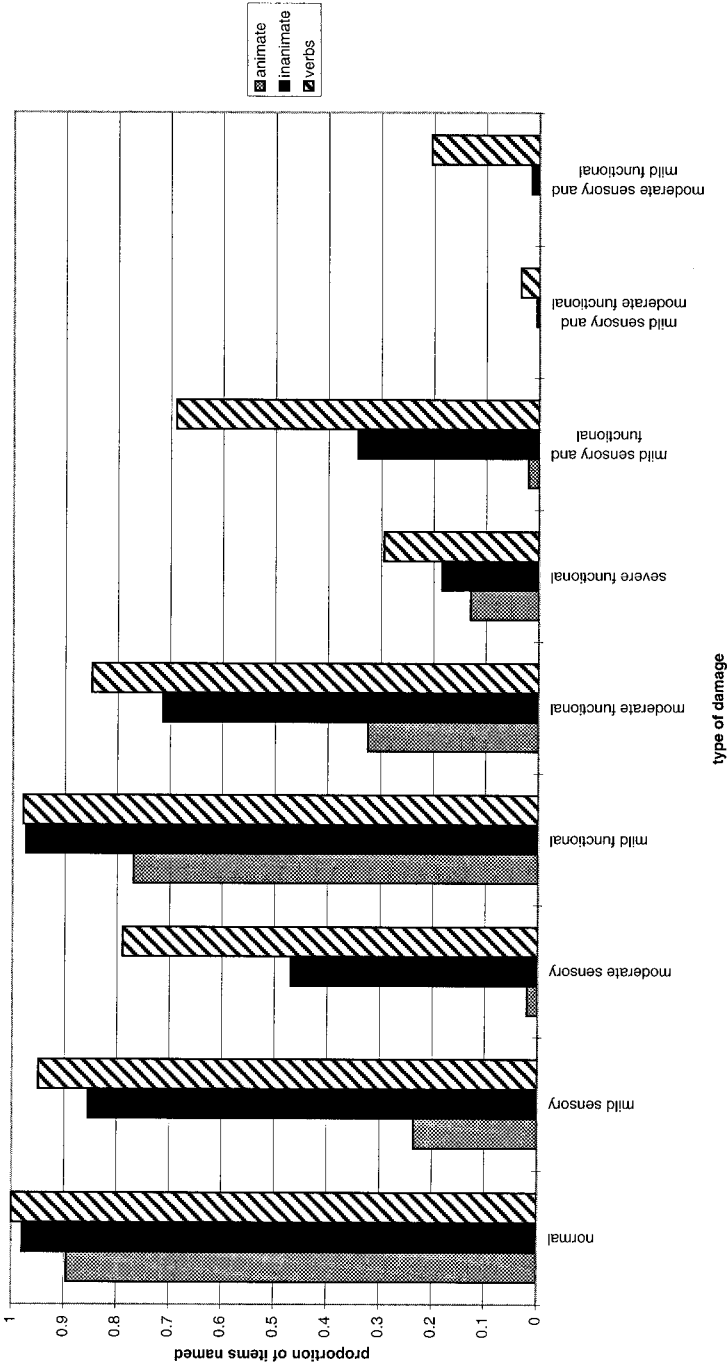


FIG. 2. Model A: Effects of different lesions with imageability controlled.

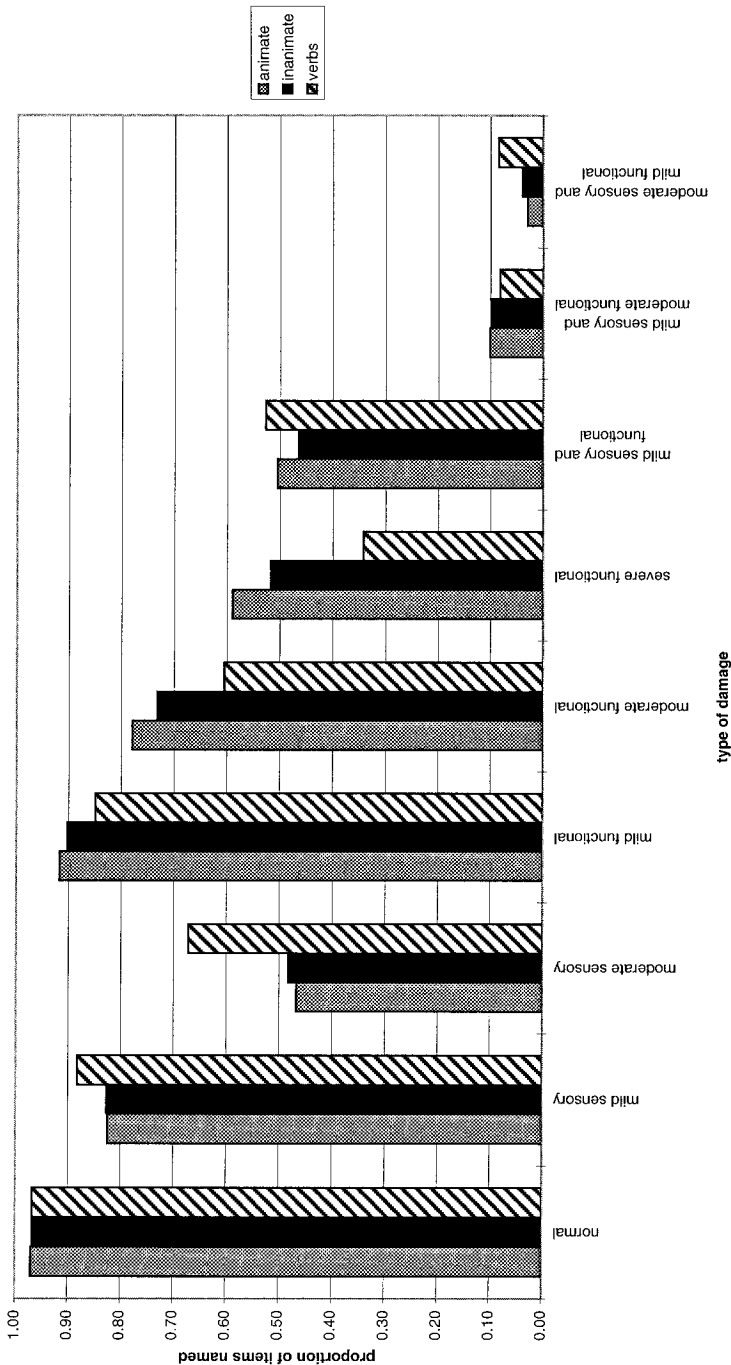


FIG. 3. Model B: Effects of different lesions.

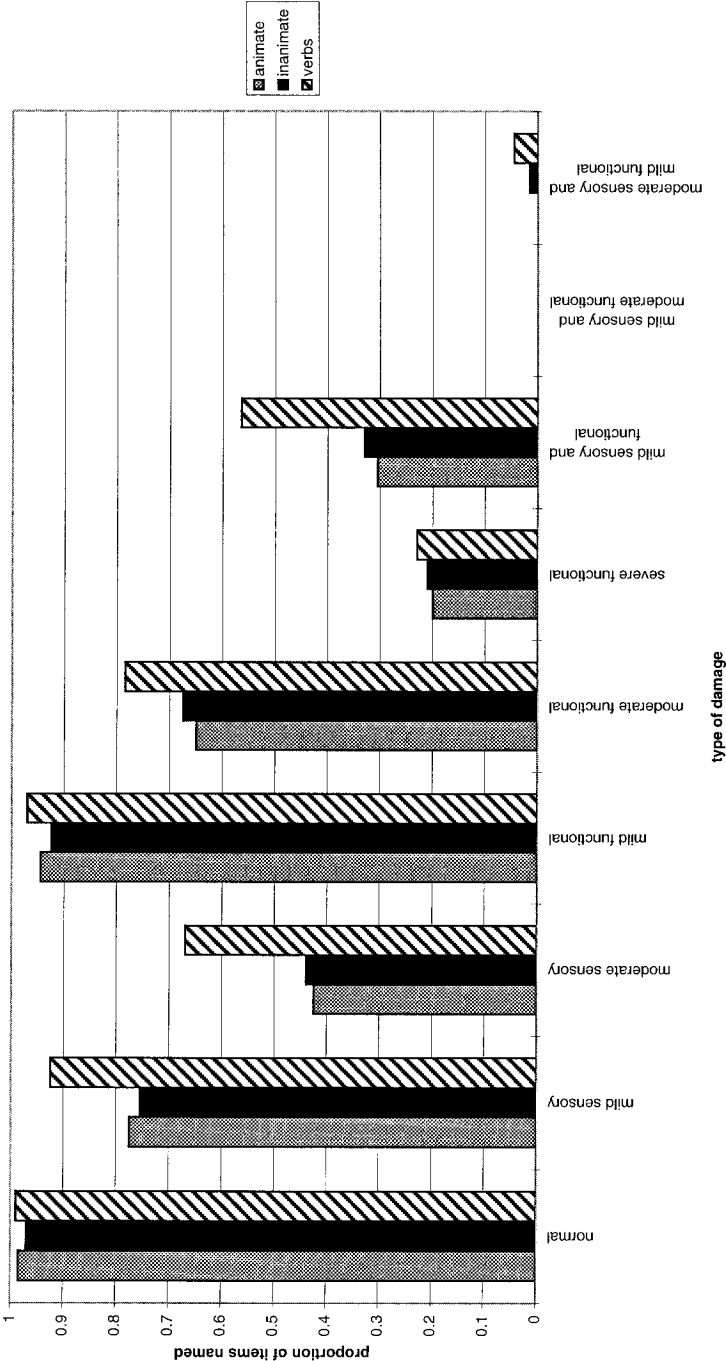


FIG. 4. Model B: Effects of different lesions with imageability controlled.

to be named; greater functional damage makes inanimates and verbs less likely to be named. Equal damage gives no category effects in the uncontrolled sets.

In both models A and C, the effects shown with sensory damage were more pronounced when imageability (number of sensory features before damage) was controlled. A controlled set would contain the inanimate items and verbs which have sensory features equal to the animates, and therefore, because of their different distribution, they will have a greater number of total features than the animates. This is why the inanimates and verbs have an even greater advantage in an imageability controlled set when sensory features only are damaged. Thus the animacy effect is exaggerated (and hence the grammatical class effect as well). Models A and C diverge, however, with functional feature damage under control for imageability. In model A, the deficit for inanimate items is reversed to give a deficit for animate items, so a mild sensory deficit looks the same as a moderate functional deficit. This is for the same reason as discussed above: the uneven distribution of feature types suffered by animates makes them more vulnerable to damage. The results (using the uncontrolled list) follow the original prediction, but controlling for the number of sensory features, while the average total features is the same for all three word types, gives animate items a considerable disadvantage throughout and verbs are always (relatively) spared.

Verbs in model C are always disadvantaged because they have fewer features to begin with (because imageability is represented by the total number of features). This means that even when there is moderate sensory damage combined with only mild functional damage, verbs are still slightly less likely to be produced than nouns (assuming equal numbers of animate and inanimate nouns) when imageability is not controlled. In contrast to model A, however, controlling for imageability does not weaken these reverse animacy effects, although normal (animates worse) effects are still increased.

Raising the threshold for naming—A strong imageability effect. The design of models C and D allows us to raise the threshold by resetting the total number of features required to access the word (Fig. 9). Doing this in models A and B would have no effect, because the mean total number of features is constant across the three word types. As model D was unsuccessful in simulating animacy effects, we confine this discussion to model C. The total number of features varies across the three word types, so it is unsurprising that raising the required number and damaging both sensory and functional features equally have exactly the same effects. Both settings render items with fewer features less likely to be activated, and hence verbs fare worst, with inanimates falling only slightly behind animates. More interesting are the effects when the raised threshold is combined with sensory or functional feature damage. This decreases the probability of finding verbs better named than nouns even in the face of sensory feature damage. An increased thresh-

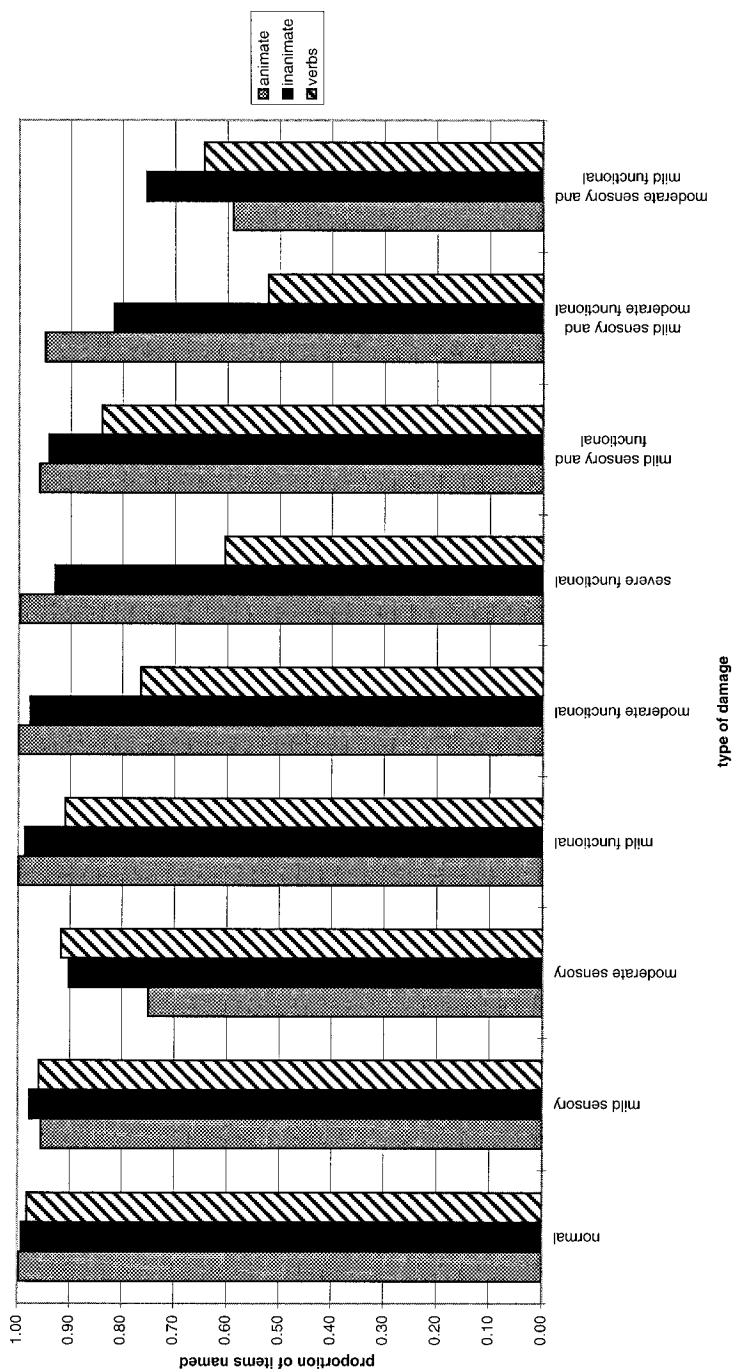


FIG. 5. Model C: Effects of different lesions.

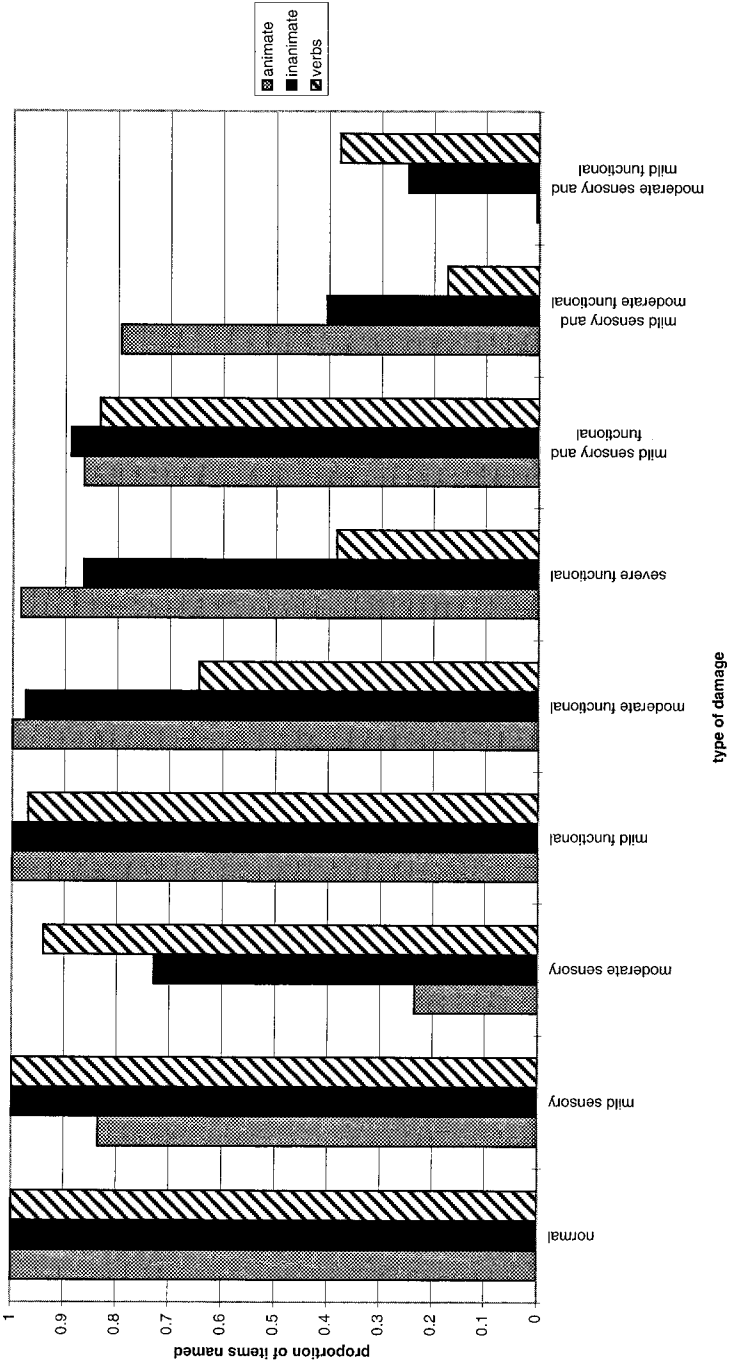


FIG. 6. Model C: Effects of different lesions with imageability controlled.

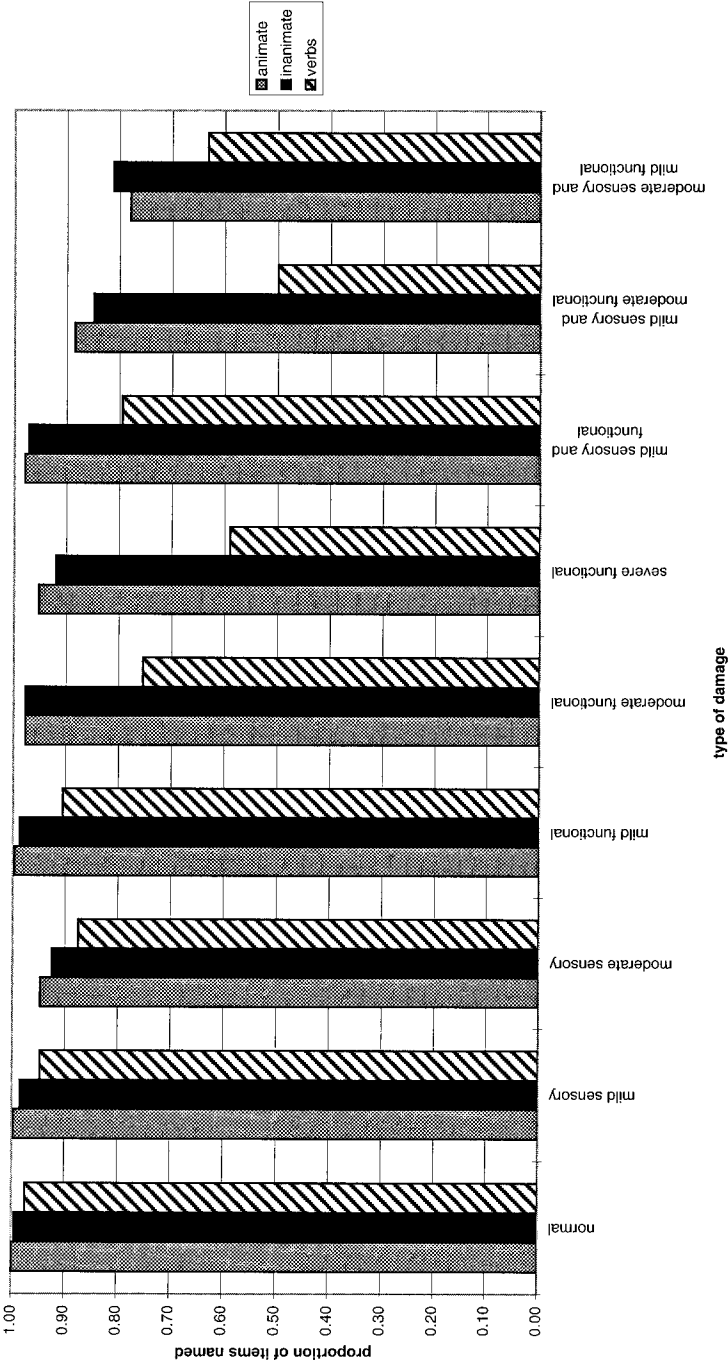


FIG. 7. Model D: Effects of different lesions.

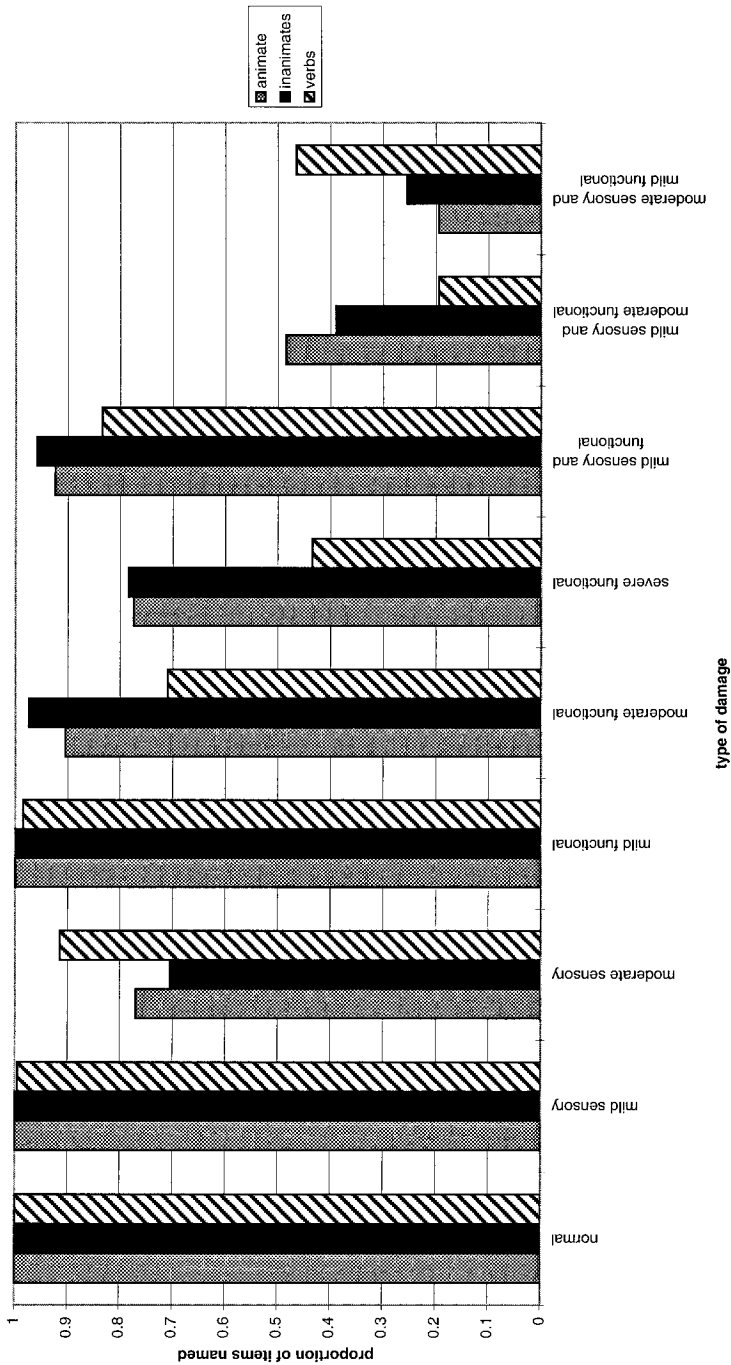


FIG. 8. Model D: Effects of different lesions with imageability controlled.

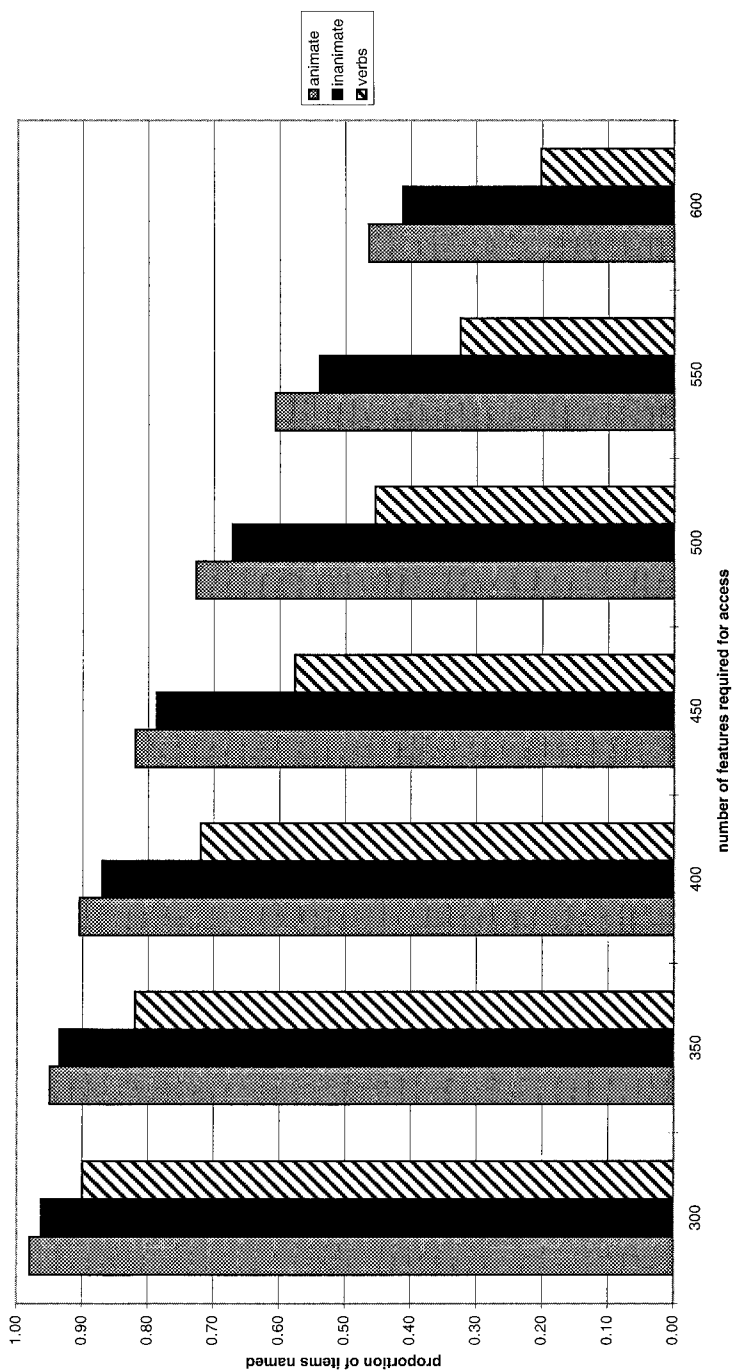


FIG. 9. Model C: Effects of raising the threshold.

old, accompanied by mild sensory feature damage, results in a slight disadvantage for animates, but no overall difference between noun and verb naming. This is a similar pattern to that which results from a combination of mild functional and moderate sensory feature damage (with no control for imageability), but the effects of a raised threshold with sensory feature damage are less drastic (Fig. 10).

Discussion

Models B and D did not produce animacy effects such as we see reported in the literature. This was because the difference in the number of sensory features between the two semantic categories was too small. In model B the number of sensory features used were the actual mean imageability ratings for these categories of nouns taken from Howard et al. (1995); in model D these same figures were used as the total number of features. Neither of these models were satisfactory in imitating the effects that have been reported.

Models A and C did produce animacy effects and grammatical category effects in both directions, all of which have been reported in the literature. The common pattern to emerge from all manipulations of these models is the co-occurrence of noun deficits with animacy effects (animates worse) and verb deficits with reverse animacy effects: the ratios of correct inanimate to animate items and the ratios of correct verbs to nouns correlated positively. This demonstrates that, in spite of the variation in methods used to represent the data, a consistent prediction emerges: a deficit for animates in relation to inanimates (if it is indeed due to damage to sensory features) is likely to accompany a deficit for nouns in relation to verbs. We have seen that a noun/verb deficit in this direction is only likely when the threshold for the number of features has not been raised, so when a strong imageability effect is seen in a patient, it is more likely that verbs will be disadvantaged because of that. For this reason the prediction does not work in reverse: a deficit for nouns will not necessarily be present when animates are less well named than inanimates.

Our models have also shown that animacy effects with poorer naming of animates than inanimates are more likely to occur than the reverse. The literature presents more occurrences of this type of effect than the reverse. The models also predict that when a reverse animacy effect is shown in a patient, it should be accompanied by a deficit for verb naming. Again, however, the reverse is not predicted: a verb deficit might be produced by a simple imageability effect, in which case we would not expect to see significant animacy effects (though perhaps a slight trend for an animate advantage in uncontrolled sets). If a verb deficit is the result of a functional lesion, we might see a reverse animacy effect, but as we have seen, the damage needs to be quite severe to impair the inanimate objects. Few published studies have combined investigation of semantic and grammatical category deficits,

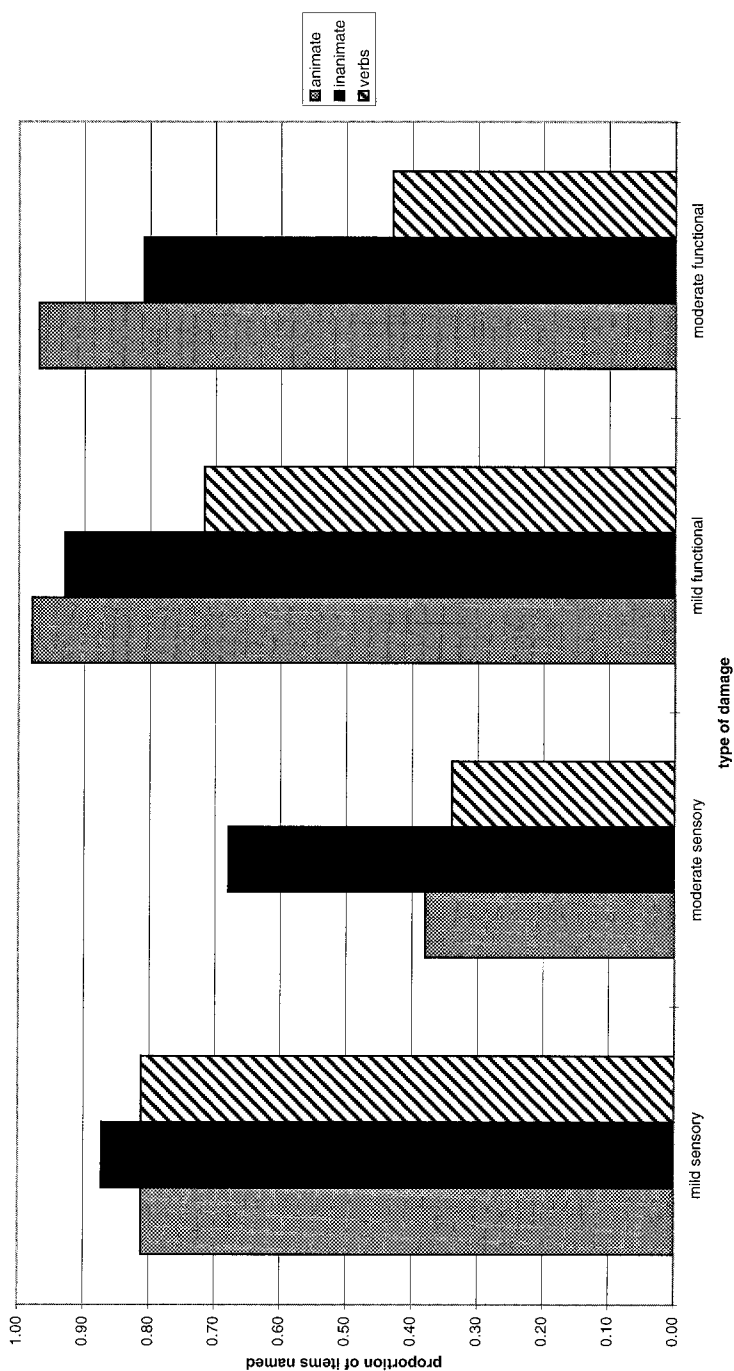


FIG. 10. Model C: Slightly raised threshold (300 features) with sensory or functional damage.

but as we discussed in the Introduction, we have been unable to find any cases in the literature on semantic or grammatical category deficits which contradict our model's predictions for the co-occurrences of the two types of deficit.

There were some potentially interesting predictions regarding animacy effects when imageability is controlled. In model C, animacy effects were slightly stronger, as animates were more disadvantaged by the uneven distribution which they have of sensory and functional features, but reverse animacy effects were not eliminated. In model A, however, no reverse animacy effects were possible when imageability controlled sets were used. The difference between the models is in the way in which imageability is represented, by number of sensory features alone (A) or by the total number of features (C). Which we choose is a matter of debate: it is unclear precisely what imageability represents in real semantic representations. This cannot easily be resolved by turning to the animacy literature. Early case studies were not concerned with control for confounding variables, until papers such as those of Funnell and Sheridan (1992) and Stewart et al. (1992) were published. These studies demonstrated that apparently clear category-specific deficits could be eliminated under control for familiarity, frequency, and visual complexity. This challenged many subsequent studies to control for these variables, and some robust animacy effects were still reported (Farah, Meyer, & McMullen, 1996).

Imageability, however, had not been suggested as a confounding factor until the paper by Howard et al. (1995). These authors demonstrated using regression analyses that many animacy effects disappear when other variables, including imageability, are taken into account. Two of the 18 patients in their study (FS and MC) initially were significantly better at naming living things, and 1 patient (GG) showed a substantial trend in the opposite direction. When their performance on matched lists (controlled for all relevant variables including imageability) was investigated, 2 patients' animacy effects disappeared (MC and GG), but 1 patient (FS) was still significantly better at naming animates. Using the controlled set, the difference between the proportions correctly named was slightly larger, but the *p* value was lower because the controlled set had fewer items. This result, however, would not be predicted by our model A, which does not allow reverse animacy effects when imageability is controlled. Howard et al. (1995) then did a regression analysis, in which all the other variables were partialled out. This showed that FS was still significantly better at naming animates, and the animacy effect shown by GG in the unmatched (but not in the controlled subset) reemerged.

Imageability was also a concern in Lambon Ralph et al. (1998) in their two case studies (DB and IW). The poorer naming of animates shown by DB was still present under control for imageability in matched sets of items.

This was a large effect, but the effect size varied between the two unmatched lists, so it is difficult to say whether it became greater under imageability control. The reverse animacy effect attributed to IW, however, was not apparent in any of the controlled lists. The only individual set on which IW was significantly better at naming animals was the 100-item (unmatched) list. The reported reverse animacy effect was only derived from regression techniques. It is difficult to say, therefore, whether our model A or C is “better” in terms of its fit with published data, but as Howard et al. (1995) retained a reverse animacy effect in their patient FS under strictly controlled variables, we suggest that model C (which allows this) has the edge. Further investigation is required into both semantic and grammatical category impairments, and more care needs to be taken to control for imageability.

The remainder of the paper tests some of the predictions that emerged from the models. Patients who show specific grammatical category deficits are tested for animacy effects and for the integrity of sensory and functional features in their semantic representations.

VERB AND NOUN CONFRONTATION NAMING

Method

A test was devised using video to test retrieval of verbs and nouns, controlled for frequency and length by phonemes and syllables (Bird & Webster, unpublished). Word frequency throughout refers to data from the Celex Database (Center for Lexical Information, 1993), using the log per million (lemmas), spoken and written combined. Each noun was represented by a 5-s clip of an exemplar of the object to be named, and each verb by 5-s clip of the action being carried out. For class ambiguous items such as *whistle*, the noun would be a static example of the object, and for verbs the action clip would not include the object with the same name, so for the verb *whistle*, a whistle was not shown, but an individual whistling using his lips. Twenty-one people with no neurological damage were tested on the clips for naming agreement, and any that achieved less than 90% agreement were discarded. Another group of 17 control subjects aged 49–69 were tested on the final version, to yield normal scores for comparison with patients. They scored a mean of 50.29 (SD 2.52) verbs and 51.59 (SD 1.28) nouns correct of 54. This trend for better naming of nouns than verbs does not quite reach significance in a two-tailed *t* test ($t(df\ 16) = 1.78, p = .09$).

A further 25 non-brain-damaged subjects rated all the items for imageability. Published imageability ratings were not suitable in this case: many items on the list had no available ratings, and even where ratings existed for items in the test, further problems regarding the different word classes arise. Those collected by Paivio, Yuille, and Madigan (1968) were nouns only, by Klee and Legge (1976) are for concreteness of verbs only, and ratings of imageability of nouns and verbs together, such as those of Gilhooly and Logie (1980), are also unreliable when the aim is to contrast the two classes, for reasons cited above. In order to achieve ratings which were truly for the noun form or the verb form as required they were given in the format “a whistle” and “to whistle.” This manner of rating resulted in no overlap in imageability between nouns and verbs used in the test: *all nouns were rated more imageable than the verbs*. This meant that controlling for imageability across pictureable verbs and nouns

was not possible, since nouns matched to verbs for imageability would not be sufficiently pictureable for use in a confrontation naming test.

Results

The six patients and their VAN scores are shown in Table 3. Also shown are these six patients' results in the verb and noun picture naming test developed by Rita Berndt and colleagues (Berndt et al., 1997). The verb deficit patients show a clear difference in the same direction as the VAN results, as does the verb spared patient ML. JS was marginally better at verbs than nouns, but NT marginally worse at verbs. In view of the rarity of true "noun deficit" patients, JS and NT were included on the basis that they fared slightly worse at nouns on one of these two tests. Qualitatively, the same can be said of NT (and indeed ML) as above regarding JS's errors on nouns and verbs. All three present as individuals who experience word finding difficulties in varying degrees, but do not show the common pattern (shown not only in most patients, but also in the normal controls) of nouns' being named more accurately than verbs. To test the assertion that these six patients fall into two distinct groups on the basis of the two tests of verb and noun naming, a combined *S* test (Leach, 1979) was used to calculate a combined *z* score for the two tests for each patient. These are also shown in Table 3. There is no overlap between the two sets of combined *z* scores, indicating that the patients do indeed form two distinct groups. The combined *z* score of the verb deficit group is -3.88 and that of the verb spared group 0.52 .

Table 4 shows further details about the six patients in this study. JS has been included in a previous study (Lambon Ralph, Sage, & Roberts, 2000) of his anomia, and his word finding difficulties were comparatively more severe at that time. IB was included in Bird and Franklin (1996) and her deficits remain unchanged since that time. Table 5 shows the results of neuropsychological tests of semantics for these patients.

That all three verb spared patients are fluent is not surprising. This goes along with classical accounts of anomia seen in fluent patients and typically agrammatism (with particular difficulties in verb retrieval) in nonfluent patients (Zingeser & Berndt, 1990). Further evidence for dividing these patients into two groups is in the mean scores for nouns and verbs. The combined average score for nouns for IB, JM, and TJ is $47/54$ on the VAN and $46/60$ on the Berndt test. The average scores achieved by JS, ML, and NT for nouns are very similar, $46/54$ and $47/60$, respectively. It is their verb retrieval which differs: for IB, JM, and TJ the mean scores are $32/54$ (VAN) and $15/30$ (Berndt); for JS, ML, and NT they are $48/60$ and $25/30$, respectively. Both groups have a naming deficit, which is of a similar magnitude for pictureable nouns, but the first group is particularly impaired at naming pictureable verbs, while the second is relatively spared for verbs. For this reason we do not refer to JS, ML, and NT as noun deficit patients, but verb spared.

TABLE 3
Patients' Results on VAN and Berndt Verb and Noun Naming

	VAN			Berndt			Combined z score
	% correct verbs (<i>N</i> = 54)	% correct nouns (<i>N</i> = 54)	Fisher exact (two-tailed <i>p</i>)	% correct verbs (<i>N</i> = 30)	% correct nouns (<i>N</i> = 60)	Fisher exact (two-tailed <i>p</i>)	
“Verb deficit” patients	46	81	<i>z</i> = 3.59 <i>p</i> = .0003	27	65	<i>z</i> = 3.19 <i>p</i> = .0014	<i>z</i> = -3.10
	74	98	<i>z</i> = 3.32 <i>p</i> = .0009	77	88	<i>z</i> = 1.12 <i>p</i> = .2607	<i>z</i> = -1.43
	57	83	<i>z</i> = 2.73 <i>p</i> = .0064	47	77	<i>z</i> = 2.59 <i>p</i> = .0095	<i>z</i> = -2.26
“Verb spared” patients	83	96	<i>z</i> = 1.90 <i>p</i> = .0574	93	92	<i>z</i> = -0.14 <i>p</i> = .8899	<i>z</i> = -0.53
	93	72	<i>z</i> = 2.52 <i>p</i> = .0119	87	70	<i>z</i> = 1.47 <i>p</i> = .1426	<i>z</i> = 1.45
	89	87	<i>z</i> = 0.00 <i>p</i> = 1.00	73	75	<i>z</i> = 0.09 <i>p</i> = .9323	<i>z</i> = 0.04

TABLE 4
Patient Details

	Sex	Age	Occupation	Etiology	Years postonset	CT scan	Fluency
IB	F	49	Shop assistant	CVA	7	Subarachnoid hemorrhage from a left middle cerebral artery aneurysm and infarct	Nonfluent
JM	F	61	Psychiatric nursing sister	CVA	2	Left middle cerebral artery aneurysm, left temporal hematoma	Nonfluent
TJ	M	56	University lecturer	CVA	2	Left frontal intracerebral hemorrhage	Nonfluent
JS	M	42	Businessman	Head injury	2	Large low density area in the left temporal-parietal region, wedge-shaped low density area in the left temporal lobe	Fluent
ML	F	67	Housewife and mother	CVA	1.5	Large wedge-shaped infarction in the left temporal-parietal occipital region	Fluent
NT	M	67	Telegraphist for a local newspaper	CVA	2	Technically poor due to movement, no information as to localization of lesion	Fluent

TABLE 5
Results of Semantic Tests

	PPT ^a (<i>N</i> = 32)	Shallice and McGill ^b concrete (<i>N</i> = 30)	Shallice and McGill abstract (<i>N</i> = 30)	Shallice and McGill emotional (<i>N</i> = 15)
Normal ranges	49-52	27-30	21-29	11-15
IB	49	28	13 ^c	9 ^c
JM	48 ^c	30	23	13
TJ	48 ^c	28	25	12
Mean verb deficit group	48.33	28.67	20.33	11.33
JS	51	30	27	15
ML	47 ^c	27	25	8 ^c
NT	46 ^c	28	14 ^c	8 ^c
Mean verb spared group	48.00	28.33	22.00	10.33

^a PPT, Pyramids and Palm Trees (pictures only version) Howard and Patterson (1992).

^b Shallice and McGill concrete and abstract word to picture matching (auditory version), Shallice and Coughlan (1980).

^c Indicates a score outside normal range.

The results of the VAN for each patient were entered into a logistic regression analysis (stepwise), and imageability was shown to be a significant predictor of naming success in all patients except NT (other variables were also significant for some patients, but we will not consider those here, see Table 6). Imageability is confounded with word class, verbs all being less imageable than nouns, and class did not emerge as a significant predictor for any of the verb deficit patients. Furthermore, in partial correlations, no significant effects were found of word class when imageability was partialled out, except for ML ($r = .258$, $p = .005$). Significant effects of imageability were found when word class was partialled out for IB, JM, and JS (all $ps < .05$). This suggests that an apparent verb deficit might be an artifact of the effect of lower imageability. The normal control subjects, as a group, show the same effect, which suggests that the patients in the verb deficit group show the "normal" pattern, but it is exaggerated due to their neurological damage. This finding represents an important challenge to previous studies reporting "verb deficits," while not controlling for imageability. It is unfortunate that, due to the lack of overlap in imageability ratings of pictureable nouns and verbs, controlling for imageability in confrontation naming seems an impossible task.

Turning now to the verb spared patients, correct naming by both JS and ML was also predicted in part by imageability. Word class was the most significant predictor for ML, however, and this was not significant for any other patient. The effect of imageability in ML is in the "normal" direction, so her class effect is not due to better performance on lower imageability items. She seems to present a normal imageability effect, but accompanied by a class-specific deficit for nouns.

Discussion

Normal subjects as a group also named fewer verbs than nouns. This might be because actions are more likely to have alternative verb "names," but all items with less than 90% naming agreement were removed from the test. It is possible, and we would suggest very probable, that many of the verb deficit cases reported are worse at naming verbs only because verbs are less imageable and hence lack the semantic richness required to activate the word forms sufficiently for production. This seems to be the case in our verb deficit patients: we have demonstrated by the regression analysis that imageability is a very important factor, and no independent effect of grammatical category is shown in this patient group. If verbs are less likely to be produced than nouns because they are lower in imageability, we would not expect to find an effect of animacy in the verb deficit group. If they are better at naming pictures of animate items than inanimate, then this would suggest that they also have degraded functional feature information, which might then be causing their poor verb naming. If the problems naming concrete nouns experi-

TABLE 6
Regression Analyses on VAN Results

[illegible]

enced by the verb spared group are due to sensory feature damage, then we would expect them to be less successful in naming pictures of animate than inanimate things.

NAMING TESTS MANIPULATING ANIMACY

All the patients described above experience word finding difficulties which result in poor performance on confrontation naming tasks. We have suggested that the lower imageability of verbs compared with nouns caused the grammatical category of verbs to appear more severely affected in three patients (IB, JM, and TJ). In the other three patients, JS, ML, and NT, retrieval of verbs seemed to be spared relative to nouns. This is not because there is any reverse imageability effect to be seen in JS, ML, or NT. We have suggested instead that these verb spared patients experience difficulties producing concrete nouns, because the sensory features on which these words particularly depend are damaged in these patients.

The models indicated a relationship between deficits for verbs and inanimate objects, because, we argue, they both differ from animate nouns in their lack of sensory features relative to functional features. The models predict that a patient who is significantly worse at nouns than verbs in confrontation naming, such as ML, should also be worse at naming living things compared to nonliving things. Worse performance at verb naming, however, does not necessarily predict the reverse animacy effect. The prediction is only in the opposite direction: when inanimate objects are worse than animates, we should expect verbs to be severely affected. According to these models, by looking for animacy effects, we might be able to differentiate poor verb naming due to a functional feature deficit from poor verb naming due to difficulties accessing low imageability words. If the latter is true for our verb deficit patients, we would not expect any effect of animacy to be present.

Method

All six patients were asked to name three sets of pictures, each of which contained equal numbers of animate and inanimate items. The first set (Gainotti & Silveri, 1996) comprises 60 items, covering a range of familiarity ratings. Their animate set contains fruit and vegetables as well as animals. Set 2 (Garrard, Patterson, Gomez-Anson, & Hodges, submitted for publication) was originally designed as a test of comprehension and comprises 80 items, grouped into eight categories. Set 3, described in Lambon Ralph et al. (1998), consists of 32 animals (no fruit and vegetables) and 32 man-made objects (no musical instruments). The latter set is the only one which is properly controlled for frequency, length, imageability, familiarity, and age of acquisition. The scores on all these tests for the six patients are shown in Table 7.

Results

The results demonstrate that the patients who are relatively spared at verb naming compared with noun naming are also consistently worse at naming

TABLE 7
Patients' Results (Percentages) on Animacy Naming and Ratio of Inanimate/Animate Pictures Correctly Named

	Set 1		Set 2		Set 3		Combined z score	Ratio inanimate/animate
	Animate (N = 30)	Inanimate (N = 30)	Animate (N = 40)	Inanimate (N = 40)	Animate (N = 32)	Inanimate (N = 32)		
IB	50	70	60	40	72	63	0.63	0.92
JM	87	93	83	83	88	94	-0.27	1.05
TJ	87	67	65	55	81	69	1.14	0.83
Mean verb deficit patients	74	77	69	59	80	75	0.80	0.93
JS	67	93	78	78	84	97	-0.82	1.15
ML	67	77	45	70	47	66	-1.77	1.36
NT	53	80	50	90	63	72	-2.37	1.48
Mean verb spared patients	62	83	58	79	65	78	-2.81	1.33

animate items than inanimates. Individuals' test results do not show significant differences in their abilities to name animate and inanimate items. There is, however, a clear trend when the three verb spared patients are combined. This pattern does not apply to the three verb deficit patients, so it is not an artifact produced by animals' being more difficult to name. Their pattern is less consistent: while IB and TJ were better at naming animates, JM was very slightly worse. As a group, they are better at animate items. When the combined *z* scores are calculated, using the combined *S* test as for the two verb and noun naming tests (Leach, 1979), there is again no overlap between the two groups.

A correlation of the ratios of inanimate to animate naming, with those of verb to noun naming, gives $r = .846$ ($p = .04$). It appears that the relative difficulty in producing either verbs or nouns is a predictor of the difficulty in producing inanimate or animate items. It is important to note, however, that this applies here only for confrontation naming.

Discussion

Why should there be a correlation between naming of a particular grammatical category and naming of a semantic category? We have proposed that both effects are due to a common underlying impairment. We have suggested that the grammatical category deficit exhibited by the verb deficit patients is in fact due to verbs' being of lower imageability, but that a reverse imageability effect cannot account for the pattern shown by verb spared patients. For the verb deficit patients, whose naming is primarily predictable by imageability, one would assume that there would be no difference in performance across the semantic categories in an animacy naming test controlled for imageability. As a group, the verb deficit patients do show little difference on the imageability controlled (set 3) test, although both IB and TJ were better at naming the animals, but only TJ shows a consistent preference for animates in all three tests.

None of the three verb spared patients, in any of the three tests, fared better at animates than inanimates. In fact, there is only one example (JS on set 2, in which he scored 78% for both categories), in which an individual in this group was not worse at naming the animate items. This cannot be explained by imageability, frequency, length, or age of acquisition, as set 3 (Lambon Ralph et al., 1998) is controlled for all these variables. It might, however, be due to the difference in the structure of the semantic representations of these categories of items. The weighting of sensory features must be somewhat confounded with imageability, as we might expect items for which sensory (especially visual) features are important to arouse a sensory mental image more easily. The verb spared patients' pattern in the noun-verb and the animate-inanimate dimension, however, shows that there is more at work here than imageability.

If the nature of semantic representations lies behind our verb spared pa-

tients' difficulties in retrieving specific lexical items, is this due to a central semantic deficit affecting sensory features, and hence affecting their comprehension of these items, or is it affecting some kind of output lexical semantics? None of the patients revealed any problems in spoken word to picture matching, using concrete nouns, both animate and inanimate.² It might be assumed from these results that they have no central semantic impairment. The results of the nonlinguistic Pyramids and Palm Trees test (Howard & Patterson, 1992) do not show any serious semantic impairments, and the word to picture matching for more abstract words (Shallice & Coughlan, 1980) demonstrate some problems with abstract words only for IB, ML, and NT. No semantic impairment can be seen for JS; he performs within normal limits on both of these tests. The impairments in word to picture matching are only demonstrated using very abstract items, and it is difficult to see how this could affect concrete nouns.

It could be argued that word to picture matching tasks and object recognition are low level tests of concrete word semantics. Perhaps the full range of sensory feature representations are not required to achieve recognition or to match a word to the correct picture, or perhaps the presentation of pictures in these tests helps to support the damaged sensory information in the semantic representations of the verb spared group. If our verb spared patients have a more subtle semantic deficit with particular regard to sensory as opposed to functional semantics, a more sensitive test is required. The following definition production task was used in order to tap the semantics of the same items as used in one of the animacy naming tests. If our original hypothesis is correct, we should find that the verb spared patients are poor at producing sensory feature information in a task designed to elicit their semantic knowledge of concrete nouns.

PRODUCTION OF DEFINITIONS

This experiment is similar to that described in Hodges et al. (1996), with some changes in the analysis. These authors asked 10 normal control subjects and 51 patients with dementia of the Alzheimer type (DAT) to give definitions to the spoken word form of a variety of objects from various semantic categories, including living and nonliving things. Their method of analysis included counts of the information given regarding both physical and associative features. These were both split into general and specific information, but as little general information was given by either normal controls or DAT patients, these categories were combined in the final analysis. Also included in their analysis was information about superordinate category labels, total correct information units, intrusion errors, judgmental comments, and vague

² JS and ML were normal at object recognition from the BORB (Riddoch & Humphreys, 1992); JS data from Lambon Ralph et al. (2000); NT was not tested.

or irrelevant information. They also considered whether the definition had captured the “core concept.”

The present experiment was designed to test the hypotheses that individuals classified here as verb deficit patients would find giving functional descriptions more difficult than sensory descriptions and that verb spared patients would show the reverse effect. It was assumed that there would be great variation in the quantity of information provided in definitions and that output from the nonfluent patients (hence verb deficit patients) would be particularly impoverished. It was not considered appropriate therefore to use an analysis which involved the number of features described, but rather one that depended on the proportion of one type of feature to another. For this reason the Hodges et al. (1996) analysis was simplified to include a scale of functional–sensory information, similar to their counts of information units pertaining to associative and physical detail, but combining the two on one sliding scale. The other types of information were not considered separately, apart from the “core concept,” which is analogous to the “quality of definition” rating in the present study.

The first hypothesis was that animals would be defined by normal control subjects using a heavier weighting of sensory information (what they look like, sound like, and so on) than functional (what we do with them, what they do) and that the reverse would apply to inanimate objects. Farah and McClelland (1991) found that in dictionary definitions there was much more sensory information given for animate items than for inanimates. Hodges et al. (1996), however, did not achieve this effect in their normal control data: they found no interaction between animacy and type of information, and we will discuss this contradiction later. If it is true that the reason for difficulties with concrete noun production lies in access from semantics and that the semantics of the verb spared group is disrupted predominantly in the connections from sensory features, and not functional features, then we would expect to see from this patient group definitions which are impoverished in sensory information.

Method

The same list of 32 animals and 32 inanimate objects was used as in the third animacy naming test (Lambon Ralph et al. 1998), described earlier. This was chosen as it was controlled for imageability as well as frequency and length, a feature lacking in most other animacy tests. Fifteen normal subjects took part, aged between 63 and 75 years (mean age 68.47). The three verb deficit patients (IB, JM, and TJ) and three verb spared patients (JS, ML, and NT) also took part.

Subjects were asked to give definitions to the single word stimuli. Their instructions were, for each word, to give a definition which was clear but concise. They were asked to try to provide a definition which was good enough to allow someone to know what they were defining, but to keep it fairly short (preferably no more than 20 words). Normal subjects wrote their definitions, but patients gave them orally, and these were recorded and transcribed.

All definitions were sorted by items, so that all subjects' definitions of *windmill* appeared

on one page, and these were given to eight naive volunteers for rating (students at University of Newcastle, who participated for course credit). They were asked to rate them for quality, that is, to what extent the definition represented something that would enable one to say what the given word was and whether the definition contained all the most salient features of the item, by giving a mark out of 10. They were also asked to rate each definition on a functional–sensory information scale numbered 1 to 7. Their instructions for this were

to assess each definition in terms of the kinds of information provided by the individual. This will be done by considering two particular aspects which can be brought into any definition: functional information and sensory information. Functional information includes details which are not derived from the senses if one were presented with the object out of context and had never heard of it before: knowledge about habitat, country of origin or where the object or animal can normally be found, what the object or animal is used for, in the case of animals, what its habits and nature are like, and general, what we might call “encyclopaedic” information. Sensory information is that which can be derived from the five senses if presented with the object or animal. Much of this is visual, but it also might include in some cases information about the feel of it, and the sound, smell or taste. The visual domain will obviously include many things which would be called sensory information: body shape and body parts for animals, size, colour, the shape and (for inanimate objects) the material of which it is made. Your assessment should reflect, on a 7 point scale, the proportion of each definition which is devoted to sensory versus functional detail.

Due to the large number of definitions given by all 21 subjects (total 1344), and the complex nature of the rating task, only 1 subject (the main rater) completed the entire set. The remaining seven “subraters” completed varying amounts of the whole questionnaire. Where the main rater and the subraters overlapped on items the means were calculated both for quality of definition and functional–sensory scale for the seven subraters. This correlated with those of the main rater (Pearson $r_1 = .74$, $p < .001$, Pearson $r_2 = .82$, $p < .001$). It was therefore decided to use the main rater’s scores for the analysis.

Results

Quality ratings. The mean quality rating for all subjects across all items was calculated, as was the mean for each patient. Means were also calculated separately for animals and inanimate items. The same was done for the ratings on the functional–sensory scale. The results of the quality ratings are shown in Table 8.

ANOVA shows no main effect of animacy ($F(1, 18) = 1.262$, $p = .276$) but a significant effect of group ($F(1, 2) = 44.44$, $p < .001$) and an interaction between animacy and group ($F(2, 18) = 3.745$, $p = .044$). Posthoc tests (Tukey’s HSD) show, for both animals and inanimate objects, that both patient groups produce significantly poorer definitions than do normal control subjects (all $ps < .001$). The two patient groups do not differ in the quality of their definitions of animate items, but the verb spared patients

TABLE 8
Mean Ratings (of 10) for Quality of Definitions for Normal Controls and Patients

	All items	Animals	Inanimates
Normal means (SD)	7.38 (1.64)	7.39 (1.61)	7.38 (1.67)
Normal ranges	6.56–8.36	6.44–9.16	6.66–8.09
IB	2.42	2.41	2.44
JM	5.02	4.97	5.06
TJ	2.41	2.91	1.91
Mean verb deficit patients	3.28	3.43	3.14
JS	5.44	5.06	5.81
ML	5.16	4.53	5.78
NT	3.47	3.22	3.72
Mean verb spared patients	4.69	4.27	5.10

produced significantly better definitions ($p = .020$) for inanimate items than did verb deficit patients (see Fig. 11).

Ratings on the functional–sensory scale (1–7). The mean ratings on the functional–sensory scale for normal controls and patients are shown in Table 9. Entirely functional definitions would receive a score of 1, entirely sensory a score of 7, so the higher the score the more sensory information was produced. Normal control subjects have higher ratings for animals than for inanimate objects, indicating that they give more sensory information for animals, and this approaches significance by subjects (paired $t(df\ 14) = 2.08$, $p = .056$), but not by items (independent samples, $t(62) = -1.27$, $p = .208$). ANOVA comparing the normal group with both patient groups (see Fig. 12) shows main effects of animacy ($F(1, 18) = 42.06$, $p < .001$) and group ($F(2, 18) = 10.33$, $p = .001$) and a significant interaction ($F(2, 18) = 13.13$, $p < .001$). Posthoc tests show that verb deficit patients do not differ significantly from controls in their definitions of animate items ($p = .105$), but their inanimate definitions contain a significantly higher proportion of functional information than do controls' ($p = .025$). Verb spared patients, however, produce significantly less sensory information than controls in both animate ($p = .010$) and inanimate ($p = .001$) definitions. The two patient groups do not differ in their proportions of sensory and functional information for inanimate objects ($p = .410$), but for definitions of animals, verb spared patients produce significantly less sensory information than do verb deficit patients ($p = .001$).

It is most interesting that the verb spared patients, while showing the pattern common to all groups of higher ratings for animals, produce significantly less sensory information than normal controls for both animals and inanimate objects. The verb spared patients also differ as a group from the verb deficit patients in their consistent lack of sensory information for animals.

We have found that normal subjects produced a greater proportion of sensory information for animals, although statistically the difference was not

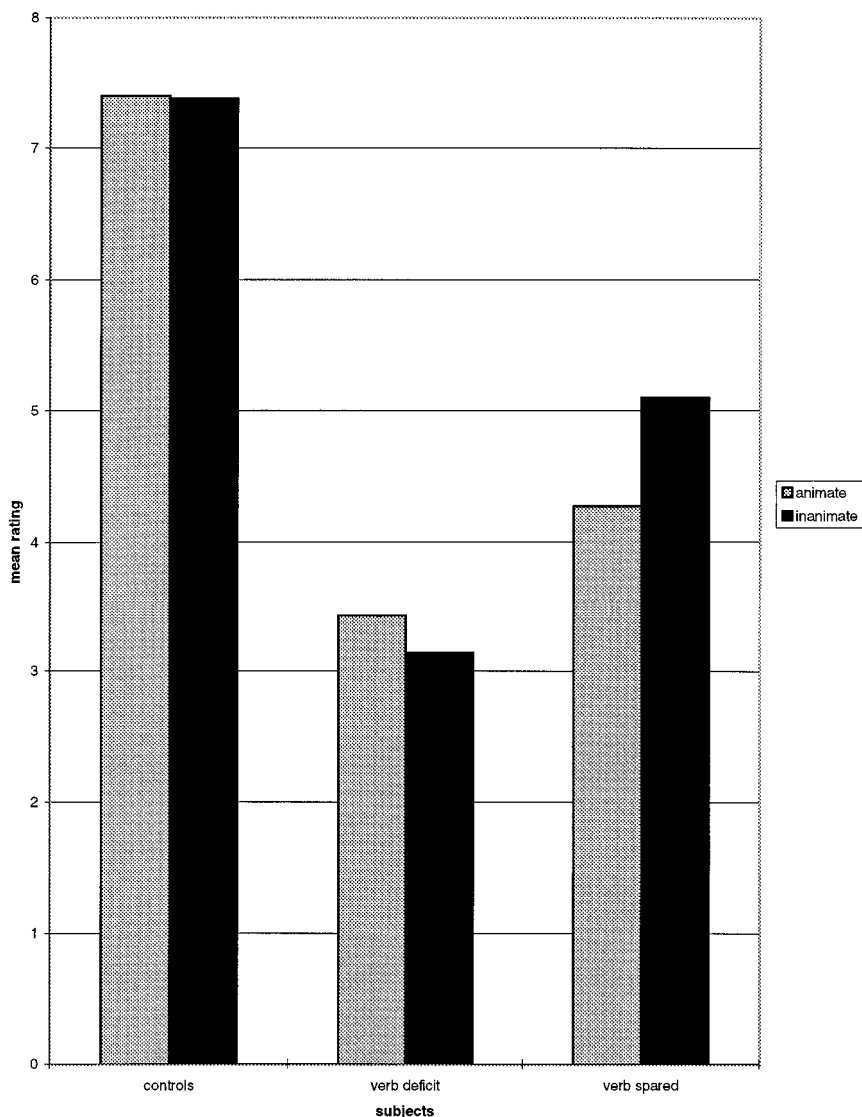


FIG. 11. Quality of definitions.

strong. Hodges et al. (1996) found no difference in their definitions test, while the difference was predicted by Farah and McClelland (1991) and the Chertkow et al. model cited in Hodges et al. (1996). The difference in results could be due to the items used. The inanimate items used by Hodges et al. (1996) included musical instruments and the present test included many

TABLE 9
Mean Ratings on the Functional–Sensory Scale (1–7) for Normal Controls and Patients

	All items	Animals	Inanimates
Normal means (SD)	3.86 (1.76)	4.03 (1.77)	3.70 (1.74)
Normal ranges	2.42–4.72	2.63–5.00	2.22–4.66
IB	4.00	5.50	2.50
JM	3.31	4.22	2.41
TJ	4.09	4.91	2.96
Mean verb deficit patients	3.80	4.88	2.62
JS	2.36	2.50	2.25
ML	2.63	3.00	2.25
NT	2.06	2.66	1.47
Mean verb spared patients	2.35	2.72	1.99

tools. It is possible that such differences could lead to their inanimates’ having more sensory descriptions than in the present study.

Why should the differences between the patient groups exist? For the verb deficit patients, the words that they find easier to produce are very highly imageable, so we might expect them to use a greater proportion of concrete nouns and adjectives, especially color adjectives. This applies particularly to IB, who is agrammatic, and TJ, who is very nonfluent, and the scores for these two patients are more sensory than those for JM, the other verb deficit patient. We might therefore expect an overall increase in sensory information, assuming that functional information requires more use of verbs, and lower imageability vocabulary. These patients do give a higher proportion of sensory information for animals than do control subjects, but this does not reach significance. They do, however, produce *less* than normal sensory information for inanimate objects. Why should the inanimate items receive definitions lacking in sensory information from patients for whom functional vocabulary is degraded? Looking at one example of definitions for the item *crown*, we can suggest that the verb deficit patients, perhaps due to their overall production difficulties, produce only the essential characteristics of each item, which in the case of inanimate objects, tend to be functional.

CROWN (normal subjects almost invariably mentioned gold, silver, and jewels, and their average score for this item was 3.2)

A typical response from the normal controls is *headgear worn occasionally by kings and queens, usually containing many jewels.* (score 3)

verb deficit patients’ responses:

- IB *queen and king Britain and monarchy and head* (score 1)
- TJ *coronation* (score 1)
- JM *prince, princess, and duchess wear the crown on their head* (score 1)

We can explain the normal but exaggerated pattern from the verb deficit patients by their selecting only the most salient features of each item, thus

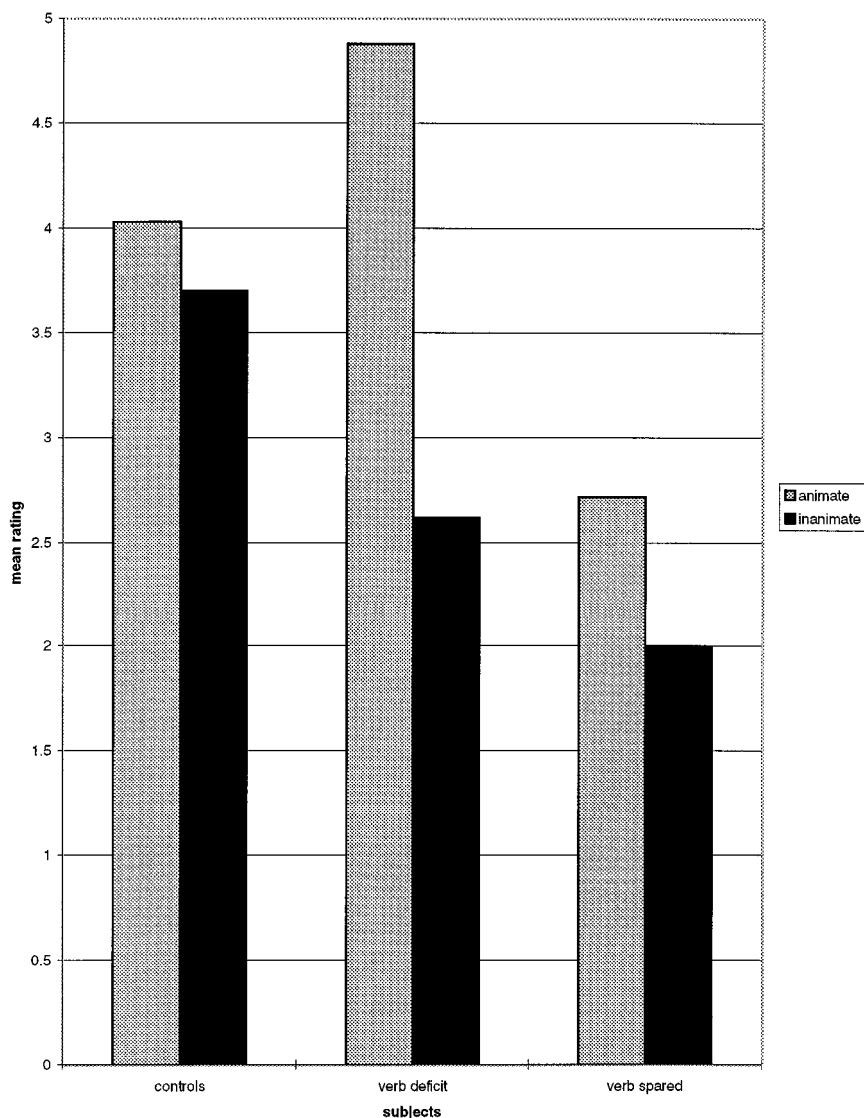


FIG. 12. Functional-sensory ratings of definitions.

making the animal definitions predominantly sensory and the inanimates' functional.

Our hypothesis for the verb spared patients was that the reason for their unusual sparing of verb retrieval was due to the sensory part of their semantic system's being more impaired than the functional. We therefore predicted that their definitions should reflect an overall tendency for predominantly

functional information, even for animals. This is borne out by the data. They show the same pattern, but a general decrease of sensory information. Two examples are shown below of a typical set of responses for animals:

- FISH (mean normal score 3.2)
Typical normal responses:
 scaly creature that lives in water, many sizes, breathes through gills and swims by using fins and tail (score 4)
 swims in sea or rivers, scaly with fins, breathes under water (score 3)
Verb deficit patients' responses:
 IB *nice, loads colors, big and small and medium* (score 7)
 TJ *bird-like, sea, fins* (score 5)
 JM *it's hot sunshine or cold fish the cold fish it's orange and about from an inch up, big one they like water, live in a tank* (score 4)
Verb spared patients' responses:
 JS *what you buy in a chip shop other than chips* (score 1)
 ML *swims in the water and very nice to eat* (score 1)
 NT *for frying and eating* (score 1)
FLY (mean normal score 5.0)
Typical normal responses:
 flying insect which buzzes in flight, transparent wings (score 6)
 small flying insect which buzzes around food (score 4)
Verb deficit patients' responses:
 IB *little little and loads legs, and little wings* (score 7)
 TJ *animal, buzzes* (score 7)
 JM *very small it's got wings, dark, and they fly* (score 6)
Verb spared patients' responses:
 JS *caught by a spider* (score 1)
 ML *little pest, flies about, onto your dinner if you let it, makes a pest of itself and my husband kills them whenever he can* (score 1)
 NT *horrible little creatures* (score 1)

It is possible that differences in the types of information provided by patients are due to an inability to access words of low frequency. The nouns and verbs used by each individual in the definition production task were extracted (by type) for each definition. The log frequency per million (by lemma) was retrieved from the Celex Database (Center for Lexical Information, 1993) for each of the nouns and verbs produced by three of the normal control subjects and all six patients (see Table 10).

TABLE 10
Frequency of Nouns and Verbs Produced by Three Normal Controls and the Two Patient Groups

	Nouns: mean log frequency per million (SD)	Verbs: mean log frequency per million (SD)
Normal control subjects	1.76 (.22)	1.94 (.37)
Verb deficit patient group	1.92 (.32)	2.19 (.58)
Verb spared patient group	1.97 (.30)	2.33 (.50)

A one-way ANOVA showed no difference between the three normal subjects. The $3 \times 2 \times 2$ ANOVAs by subjects and by items revealed a main effect of class ($F1(1, 6) = 6.23, p = .047, F2(1, 56) = 27.35, p < .001$), nouns being on average of lower frequency than verbs. A main effect of subject (normal, verb deficit, and noun deficit) was apparent not by subjects but by items ANOVA ($F1(2, 6) = .93, p = .445, F2(2, 112) = 16.10, p < .001$). There was no main effect of animacy of stimulus (whether the words produced were for an animal definition or for an inanimate object), nor were there any interactions between any of these factors. In $2 \times 2 \times 2$ ANOVAs comparing the two patient groups, only class was shown as a significant main effect, and only in the by items analysis ($F1(1, 4) = 3.72, p = .126, F2(1, 57) = 22.76, p < .001$). No effect of group, animacy, or any interactions were shown. We can conclude from this that while the patients do tend to produce higher frequency items, this is the case for both nouns and verbs and across both patient groups. We cannot account for the patient group differences on the functional–sensory scale by appealing to a word frequency effect.

Discussion

The list of words used in the definition production task was controlled for length, frequency, familiarity, age of acquisition, and imageability. It was hypothesized that in spite of the homogeneity of the items' imageability ratings, the animals would prompt definitions from normal subjects which were more heavily weighted for sensory features than those for inanimate objects. This was borne out by the data, which provide support for our suggestion that imageability ratings are not simply a measure of the number of sensory features in a semantic representation. All individual patients showed the same pattern of greater sensory information for animals as well. If the animate items, matched for imageability with inanimate items, attracted a greater weighting of sensory features than the inanimates, this implies that distributions of sensory features and imageability ratings are not measuring the same thing. The assumption in model A, but not in model C, was that the number of sensory features was directly measured by the imageability ratings. We would suggest therefore that model C, which used imageability ratings to derive the total number of features (sensory and functional combined), is the closest out of the four in its simulation of semantic representations.

Although individual normal control subjects varied quite widely on the functional–sensory scale, we have demonstrated clearly that there is a difference between verb spared patients and normal controls and between verb spared patients and verb deficit patients. Verb deficit patients show an exaggerated effect of the preference to give sensory information for animal stimuli and functional information to inanimate stimuli. We have suggested that

this is in part due to their impoverished output, which results in their producing only information which is most salient to the item. Verb deficit patients did not lack functional information, so we can assume that their verb deficit was not due to functional feature damage, but was the result of the strong imageability effect revealed by the regression analyses on the VAN results. In contrast, verb spared patients show a decrease in sensory information across both categories. Thus the decrease in access to sensory semantic features, hypothesized to be the cause of their difficulties in access to concrete nouns, is also shown in this task.

We have seen a persistent trend in naming tasks manipulating animacy: verb deficit patients do slightly better at naming animals, and verb spared patients considerably better at naming inanimate objects. While the differences in individual cases do not reach significance, it is notable that the same trend is also found in the quality of the definitions produced. The evidence from the definition production task confirms our hypothesis that a correlation would be seen between sparing of verb retrieval and sparing of functional semantic features. It is concluded that the weighting of sensory versus functional semantic information lies at the root of observed differences in the production of pictureable nouns and verbs and those between animate and inanimate nouns.

VERB AND NOUN NAMING TO DEFINITION

The results of the definition production task demonstrated that the verb spared patients had suffered degradation of the sensory features in their semantic representations. This accounts for the effect of animacy, because animate items depend on sensory knowledge to a greater extent than do inanimate concepts. We have suggested that degradation of sensory features can also account for the sparing of action verb naming relative to concrete noun naming in this patient group. The verb deficit patients did not show the reverse effect of damage to functional features in the production of definitions. Our models demonstrated that verb deficits could be due to functional feature damage, but could also be due to the effect of imageability: greater difficulties in producing low imageability words would produce an apparent verb deficit in picture naming, as verbs attract lower imageability ratings. The VAN and the Berndt naming test assessed retrieval of verbs and nouns controlled for frequency and length, but these did not control for imageability. The results of the regression analysis on the VAN suggested that the apparent specific word class deficit shown by the verb deficit patients might be an artifact of lack of control for this variable. These patients showed a very strong effect of imageability and none of word class, and, as all the verbs were less imageable than the nouns, imageability might be the reason for their apparent word class effect. A naming to definition test was designed, therefore, to control for imageability, as well as frequency and length (which

necessitated using nouns of lower imageability than in confrontation naming tasks). If the verb deficit patients' poorer performance on naming verbs is due to imageability, and not functional feature damage, a test which controls imageability should show no word class effects.

We have shown that the relative sparing of verb production in the verb spared group is not the result of a simple reverse imageability effect, but the degradation of sensory feature knowledge. We would not automatically expect, therefore, that controlling for imageability should cause the word class effect to be eliminated. Nevertheless, when we use lower imageability nouns to match the verbs, we also use items that are much less rich in sensory information (the items used are not readily pictureable). It is possible that both word classes will therefore be low in sensory information, and hence the nouns as well as the verbs will not be of the type to cause problems for the verb spared group. This "accidental control" for sensory information might make the verbs and nouns in this test equally available to the verb spared patients.

Method

A set of 25 unambiguous verbs and 25 unambiguous nouns was selected and matched pair-wise as closely as possible for frequency, length (phonemes and syllables), and imageability (matching data are shown in Table 11). Note that the mean imageability rating (452.5) is very close to that of the verbs in the verb and noun naming test (459) described earlier. Each item was given a definition taken from *Collins English dictionary* (1994), for example, *the child of one's aunt or uncle* (cousin), *to take someone as one's husband or wife* (marry).

In pilot testing, the written list of definitions was given to four normal young control subjects. Some items received quite a wide variety of responses. It was therefore decided that subjects should be given the first phoneme and first letter of the target word. Further pilot testing showed that this method provided very consistent responses. In most cases, subjects would either produce the target or a nonresponse. Although the patients might differ in their ability to use such cues, since each individual acts as their own control across the word classes, this method should not produce a class difference.

The assessment was tested auditorily on 24 normal control subjects. Twelve were aged between 61 and 69 (average 65.58) years and 12 were aged from 70 to 76 (average 72.33). Subjects were tested in two group sessions and wrote down their responses. Verbs were tested before nouns, and each set was introduced by saying they were all "action words" or "things," respectively. Each definition was read out to the subjects, and then they were told what the first sound of the word was and what the first letter was. The stimuli were repeated as many times as requested.

TABLE 11
Matching Data for Verb and Noun Naming to Definition Test

Means (SD)	Frequency	Syllables	Phonemes	Imageability
Verbs	1.22 (0.67)	1.84 (0.69)	4.64 (1.35)	453 (35.04)
Nouns	1.20 (0.68)	1.84 (0.69)	4.64 (1.22)	452 (35.07)

TABLE 12
Naming to Definition: Normal Controls' and Patients' Scores

	Verbs correct/25	Nouns correct/25	Total correct/50
Normal controls mean (SD)	20.50 (2.87)	19.79 (3.56)	40.29 (5.95)
IB	2	1	3
JM	7	5	12
TJ	5	7	12
Mean verb deficit patients	4.67	4.33	9.0
JS	16	16	32
ML	14	15	29
NT	7	9	16
Mean verb spared patients	12.33	13.33	25.67

Results

The younger group produced more correct responses (<70 group mean 42/50, range 31–50; 70+ group mean 39/50, range 29–47), but the difference was not significant, $t(df\ 22) = 1.44$, $p = .164$. The results for the two age groups were collapsed to provide normative data for comparison with neurologically impaired subjects and are shown in Table 12. The mean number of correct responses for nouns is slightly lower than that for verbs, and nouns have a larger standard deviation. These differences, however, are not significant, $t(df\ 46) = .758$, $p = .452$. The six patients were tested individually in exactly the same manner, except they responded orally. Their scores are also shown in Table 12.

This test is considerably more difficult than confrontation naming, and this is reflected both in the normal controls' and in the patients' scores. As mentioned above, the mean imageability of these items was perforce lower than that of the nouns in the VAN. None of the six patients showed any word class effect in this test. For the verb deficit group, this supports our theory that their apparent poorer performance on verbs is due to their lower imageability: when imageability is controlled, the verb deficit is eliminated. For the verb spared group, the explanation is less clear, as their apparent verb sparing is also eliminated.

Discussion

We have already ascertained that for ML, the verb spared patient who showed significantly better performance at verb naming, this was not due to a reverse imageability effect. Our hypothesis was that a deficit for nouns compared with verbs is due to damage to sensory features in semantics. The elimination of the word class effect in the verb spared patients implies that the proportion of sensory to functional features is the same across nouns and verbs in the naming to definition test. In the VAN and the Berndt tests, we

must assume that the nouns had more sensory features than did the verbs. We also know that the nouns were more highly imageable than the verbs in those tests, but that this was not the reason for the difference in performance across the word classes for ML. Perhaps then, the verb spared patients do not show a class difference here because the number of sensory features is the same across classes.

If the lack of class effect is not due to control for imageability, but due to equal proportions of sensory features, then these two factors are not the same. Two of our models (A and B) assumed that imageability was directly related to the number of sensory features, but the other two (C and D) assumed that imageability was a measure of the total number of semantic features in a representation. The data presented here lends some support to the latter. The statistical analyses of the VAN scores provided evidence that the class effect would have remained for ML, even if it had been possible to control for imageability in the VAN. There was no reason to assume, therefore, that her class effect would disappear in naming to definition. It is, however, reasonable to suppose that, when we are forced to use items which are not pictureable, we effectively lose most of the sensory features. The nouns and verbs used in the naming to definition test, we must assume, are equally lacking in sensory information, and at the same time, (because they are controlled for imageability) they have equal number of features in total. If the items used in this test depend for the most part on functional features, and assuming our verb spared patients have relatively spared functional features (and no other deficit), they should have little problem with this test. Indeed, both JS and ML perform within the normal range. The verb deficit patients performed considerably worse than the verb spared patients on this test. This group all had a strong imageability effect in noun and verb picture naming, and as we have said, the items in the naming to definition test are less imageable, so it is not surprising that this task gave the verb deficit patients greater problems.

GENERAL DISCUSSION

We have presented evidence regarding verb/noun and inanimate/animate production, together with that from broad and more detailed tests of semantic representations. The models presented at the beginning of the paper were based on assumed distributions of features for animate and inanimate nouns and verbs, specifically a heavier weighting of sensory features for animate nouns, less for inanimate nouns, and more functional for verbs. Thus the semantic representations of these three word types form a continuum of more sensory and less functional to less sensory and more functional. This simple premise led to predictions arising from lesioning the models in different ways. Damage to sensory features causes deficits for living things relative to nonliving, and provided the threshold for required activation to achieve

production is not raised and functional features are not damaged, this should also result in noun naming being impaired relative to verb naming. This is, of course, assuming that the nouns in the naming test are balanced for the two semantic categories. The verb and noun naming test described here (and the Berndt test) were developed originally to look for grammatical class deficits without taking animacy into account. Animates are underrepresented in both tests, which would mean that verb spared patients were less likely to show a significant class effect, because their naming of inanimates is relatively spared.

Further investigation of three verb deficit and three verb spared patients showed that their grammatical class deficit was not present when tested on more abstract items. Verb deficit patients performed poorly on these abstract nouns and verbs, as they have strong imageability effects, which, we believe, are the reason for their apparent verb deficit. This is because, in a picture naming test, verbs are rated less imageable than nouns, and control cannot be achieved for imageability in such a test. Verb spared patients, however, were not spared at verbs because of a reverse imageability effect. Their impairment for naming concrete nouns was due to damage to sensory features in semantic representations. Thus they were better at naming inanimate objects, which do not depend as heavily on sensory information as do animates, and for the same reason they were better at naming verbs.

Confirmation of their sensory feature deficit came from the task requiring definitions to be produced to a list of animals and inanimate objects controlled for imageability. The verb deficit patients gave impoverished definitions, but the quality of their definitions was equal for animates and inanimates, and they seemed to produce the features which were most salient (more sensory for animals, but more functional for objects). In contrast, the verb spared patients gave much poorer quality definitions for animals, and this was because they suffered a significant lack of sensory information across all items. We conclude that the pattern of spared verbs and spared inanimates shown by the verb spared group was due to damage to sensory features in semantics. The verb deficit patients did not have functional feature damage, but their pattern of deficits was the result of a raised threshold, so that a greater number of total features needed to be activated in order to access word forms. This was manifested in a strong effect of imageability, which seems better represented in the model (C) which used actual mean imageability ratings as the mean total number of features for animates, inanimates, and verbs. We suggest, therefore, that imageability, while correlated with sensory features, is more a measure of semantic richness, and the more semantic richness associated with a word, the more likely it is to be produced on demand (while all other variables are held constant). While the verb deficit patients presented here did not have specific functional feature damage, our models suggest that verb deficit patients will exist which do and that these patients should show a reverse animacy effect.

Dissociations in aphasia, such as those between nouns and verbs or phonological and orthographical output representations, have been cited as evidence for separate autonomous modules or systems. It has been shown that such dissociations can be replicated by models which do not contain autonomous components (Van Orden, Jansen op de Haar, & Bosman, 1997). The modeling described in the present research demonstrates clearly that both animacy effects and grammatical category effects can be produced in both directions, and thus double dissociations can emerge from a system with no independent modules, but simply a continuum of features on a sensory-functional scale. This calls into question the importance of the double dissociation in neuropsychology. The observation of a double dissociation gives rise to a hypothesis for which further evidence must be sought; the observation is not proof in itself.

We believe these data point to a single cause for the relative difficulties in retrieval of both grammatical and semantic categories for the patients assessed here. We have proposed the inanimate/animate distinction is in part due to weighting of functional and sensory information and in part due to the semantic representations of animate items' having features with a greater tendency to overlap or "cluster." If our assertion is true, this entails that supposed "grammatical class" effects arise at the level of semantics and not at a lexical level.

The opposite view has been expounded by Caramazza and Hillis (1991) among others. These authors account for the existence of specific word class deficits by positing organization by grammatical class at the lexical level (phonological and orthographic output representations independently). The evidence presented in support of this hypothesis is that their patient HW produced more verb than noun errors in oral production, but not in written production, while patient SJD produced more errors to verb than noun targets in written production but not oral. Greater difficulties producing verbs relative to nouns is a common pattern, and, as we expressed earlier, the confound in all previously published cases has been imageability. Caramazza and Hillis (1991) argue that the use of verb and noun homonyms shows that the difference must lie in the grammatical category, because the verb and noun forms are phonologically identical. There are two main arguments against their position. We would argue first that the difference lies in the semantic representation (compare the meaning of the noun *a crack* with the verb *to crack*), and the verb forms of noun/verb homonyms attract lower imageability ratings.³ It is most important to employ tasks manipulating verbs and nouns in a controlled manner, rather than to state simply that the subject shows no effects of imageability on other tasks and has no apparent semantic deficit.

³ Imageability ratings for a set of noun/verb homonyms used in reading and writing assessments were collected by the first author (unpublished). The mean imageability ratings of the noun and verb forms were 536 (SD 104) and 456 (SD 80), respectively.

The second argument for such deficits arising at the semantic, rather than the lexical, level was put forward by Gainotti et al. (1995), drawing on evidence from apparent grammatical category-specific disorders in semantic dementia (accepted as arising from semantic level degradation). These authors proposed that a semantic disorder can be manifested to differing degrees in different modalities, and they support this with evidence from the Daniele et al. (1994) patient RA, who suffered a progressive aphasia, first presented with a grammatical category deficit (for verbs) in written naming but not in oral naming. As her disease progressed, she showed the same poorer production of verbs in oral as well as written naming, but she was still able to give good definitions of items she could not name. A year later she had significantly worse comprehension of verbs compared with nouns and her verb naming was almost at floor. Her noun production and comprehension were not spared; they deteriorated as her disease progressed, but were always better than those for verbs. This demonstrates that RA's first symptoms were the beginnings of a semantic disorder, but her problem seemed to be restricted to the written production of verbs, a pattern which would be interpreted as a class-specific lexical deficit by Caramazza and Hillis (1991). As Gainotti et al. (1995) suggest, a mild semantic deficit could have caused the disorders presented by the Caramazza and Hillis (1991) patients HW and SJD, but if their impairment was at a semantic level, testing was insufficiently sensitive to uncover this. Specifically, the Gainotti et al. (1995, p. 262) view is as follows:

When the brain damage disrupts a semantic category only at the level of the left hemisphere (as in the case of most aphasic patients) this partial degradation provokes an inability to fully activate the corresponding lexical representations. A partly similar defect could be provoked by a mechanism of disconnection isolating the semantic representation of a given category from its lexical counterparts. This hypothetical mechanism could explain, for example, the very selective deficits in verb production reported by Caramazza and Hillis (1991).

Allport (1985) proposed that object knowledge is distributed across "attribute domains" related to the relevant modality of experience, and Farah and McClelland (1991) suggested that different types of semantic knowledge (for example, visual and functional) may be stored in anatomically distinct locations, but within one distributed system of semantic memory. Such theories, while not advocating separate semantic systems, have been used to explain semantic category specific effects, such as deficits for living things arising from damage to visual semantic features. The PET study of Martin, Haxby, Lalonde, Wiggs, and Ungerleider (1995) identified an area of increased blood flow which was adjacent to the area associated with perception of movement in the left posterior middle temporal gyrus during a verb generation task; they also found that a task requiring the generation of color names to the same stimuli resulted in activation of a separate area just anterior to that

associated with color perception, the left fusiform gyrus. Martin et al. (1995) interpret their findings as evidence that attributes defining objects are "represented close to the cortical regions that mediate perception of those attributes." Thus the production not only of color words for noun pictures (and written words), but also of verbs gave rise to activation in areas used for perception. The verb generation task, however, also yielded additional activation in the left frontal lobe: middle frontal gyrus (BA 6), inferior frontal gyrus (BA 44, 45, and 47), and Broca's area (BA 44). This suggests that the semantic representations of action verbs do have sensory features, for example, those pertaining to the perception of movement, but also functional features, which perhaps lie nearer to perisylvian regions.

Martin, Wiggs, Ungerleider, and Haxby (1996) demonstrated the involvement of visual cortex in naming animals but not tools; naming of tools gave rise to activation in the left middle temporal gyrus (as for verb generation) and thus might be a site important for stored information about visual motion associated with using objects. Damage to this region might therefore hinder access to names of tools and action verbs, while a more posterior lesion, affecting the occipital or temporo-occipital regions might result in deficits affecting the naming of living things. Verb naming is also strongly associated with the left frontal premotor cortex (Martin et al., 1995), and pathology of this region is common in cases of verb deficits (Damasio & Tranel, 1993) as well as function word deficits, as Broca's aphasia is associated with anterior perisylvian lesions (Mohr & Gautier, 1995). The verb spared patients described here certainly suffered lesions more posterior to those of the verb deficit patients (see Table 4). A deficit in the production of verbs will automatically impede sentence production, but if part of the semantic representations of verb concepts also coincides with the area which is commonly thought to be associated with grammatical encoding, then this might cause the frequent co-occurrence of these deficits. In the present study, the verb deficit patients were all nonfluent. The results of a large group study on the verb and noun naming test (Bird & Webster, unpublished), however, showed that 54/57 (95%) of patients named verbs less accurately than nouns, and only 59% of these were nonfluent. The predominance of patients who name verbs less accurately than nouns here and in the literature probably reflects the prevailing effect of imageability in brain-damaged subjects. As perisylvian lesions are associated with dysfluency, and noun deficits in the presence of spared verbs (and function words) result from lesions which typically do not involve the perisylvian region, it is unsurprising that nonfluent patients with selective difficulties in the production of nouns relative to verbs are not found.

The evidence does not support the view that specific brain structures are responsible for specific semantic or grammatical categories, but rather that neuroanatomical correlates can give this appearance, because the representa-

tions of these categories are closely linked with the mechanisms used in the acquisition of the relevant semantic knowledge. Acquisition of abstract concepts depends upon verbal information (abstract meanings are learned by association with already learned words or definitions), which therefore involves little sensory or motor input. It is no surprise then that in an ERP study, Kounios and Holcomb (1994) found that abstract words activated primarily the left perisylvian region, which is considered most central to language processing, and damage to this area results in difficulties with abstract and particularly function words (Pulvermuller, Mohr, Sedat, Hadler, & Rayman, 1996). The prevalence of deficits for living things relative to nonliving was reflected in our modeling: in order to achieve similar degrees of animacy and reverse animacy effects, greater damage to functional than sensory features was required. In the model, the increased robustness of inanimate concepts was due to their having a more equal distribution of features across sensory and functional domains. In the brain, other factors might be at work: as we discussed earlier in the paper, it is possible that sensory features are more salient or more localized. Evidence from the PET studies of Martin and colleagues suggests that purely visual knowledge might be more localized than the other kinds of knowledge such as that about movement and manipulation of objects. As Wernicke suggested, "primary functions (perceptions) alone, can be referred to specific cortical areas . . . all processes which exceed these . . . are dependent upon the fibre bundles connecting different areas of cortex" (Wernicke, 1874).

Semantic category deficits have been generally assumed to arise at the level of semantics: we propose that grammatical category deficits arise at the same level. Apparent verb deficits need to be scrutinized in more detail and in better controlled conditions, before a simple imageability effect can be eliminated: we would even go as far as to claim that true "verb deficits" do not exist, when imageability differences between verbs and nouns are taken into account. We have shown that it is possible (though rare) for noun production to be differentially impaired compared with that of verbs, but only for the most concrete items. This assertion, however, also carries a caveat: there are insufficient (if any) verbs which are as highly imageable as the kinds of nouns that are impaired in these patients, so that controlled testing is rendered impossible. The coincidence of poor pictureable noun naming relative to verbs, poor naming of animate nouns relative to inanimates, and poor production of sensory information in definitions relative to functional points to an impairment of the sensory features of semantic representations relative to functional features. This is in agreement with a model of distributed semantics over a large area of the cortex, which in the case of the multitude of so-called "verb deficit" patients might have suffered physically widespread (though varying in depth) damage. In the case of verb spared patients, we suggest that the unusual nature of their impairment might

be due to damage to a relatively discrete area of the semantic cortex which deals specifically with sensory semantic information.

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